

Requirements Traceability Matrix — AZ Phase 3 VI + C&D XSteps

AZD0543 · PN 8012441 Virus Inactivation + PN 8012475 Concentration & Diafiltration · GPF 2000 L, Process 1

1 · Section-level coverage (provable)

Recordable requirement lines mapped to an XStep	464
Reference-only lines (§1-3 summary / flowchart / SOP refs)	54
Excluded (revision history)	5
TOTAL lines extracted from both MPRs	523
Recordable coverage	100%

Proof: every process section (§4 → last attachment) in BOTH source MPRs is rendered verbatim under exactly one XStep. No recordable section is unmapped. Whole sections are rendered (not filtered), so the reusable-block keyword harvest cannot drop content — it is only a redundant view.

2 · Streamlined-Analysis reconciliation

Covered: 28 Folded: 1 Review: 1 Out-of-scope: 2

Both former GAPs resolved with no new XStep: **SoloVPE/SOLO_A280** is a sample (SOP-0107091) handled by the Sampling Record XStep; the **VPro filter-integrity** worksheet belongs to the separate Virus Filtration record (MABR-0027575 / PN 8012474), cassette integrity on FORM-0071789.

3 · RTM — every recordable line → the XStep that fulfils it

Req ID · Record · § · Section · Type · verbatim requirement · fields captured · fulfilling XStep · coverage. Colour = coverage status.

Req ID	Rec	§	Section	Type	Old BR requirement (verbatim)	Fields captured	Fulfilled by XStep	Cov.
VI-S1-001	VI	§1	Process Summary	Instr.	This master production record (MPR) describes the procedure for the Viral Inactivation and Neutralization of the AZD0543 product in the downstream process outlined in this MPR.		— reference / not recordable —	Refe
VI-S2-001	VI	§2	Downstream Process Flowchart	Fields		MABR-0027568 / PN: 8012412 / AZD0543 - Centrifuge Harvest (GPF, 2000 L Scale, Process 1) MABR-0027569 / PN: 8012440 / AZD0543 - Affinity Chromatography (DurA Cycle A50, GPF, 2000 L Scale, Process 1) MABR-0027570 / PN: 8012441 /	— reference / not recordable —	Refe

						AZD0543 - Virus Inactivation (Low pH, GPF, 2000 L Scale, Process 1) MABR-0027573 / PN: 8012472 / AZD0543 - Anion Exchange Chromatography (CaptoQ, GPF, 2000 L Scale, Process 1) MABR-0027574 / PN: 8012473 / AZD0543 - Cation Exchange Chromatography (Poros HS50, GPF, 2000 L Scale, Process 1) MABR-0027575 / PN: 8012474 / AZD0543 - Virus Filtration (VPro, GPF, 2000 L Scale, Process 1) MABR-0027576 / PN: 8012475 / AZD0543 - Concentration and Diafiltration (UF Skid, GPF, 2000 L Scale, Process 1) MABR-0027577 / PN: 8012477 / AZD0543 - Formulation and Bulk Filtration (Cryovault, GPF, 2000 L Scale, Process 1) Storage Final Sterile Filtration and Vialing (off-site fill/finish manufacturer)		
VI-S3-001	VI	\$3	Referenced Documents	Instr.	SOP-0080023: Operation of the Powerblanket BHT System		— reference / not recordable —	Refe
VI-S3-002	VI	\$3	Referenced Documents	Instr.	SOP-0080165: Equipment and Material Flow in the Gaithersburg Pilot Facility		— reference / not recordable —	Refe
VI-S3-003	VI	\$3	Referenced Documents	Instr.	SOP-0080185: WFI Point of Use Operations		— reference / not recordable —	Refe
VI-S3-004	VI	\$3	Referenced Documents	Instr.	SOP-0080294: Floor Scale Operation		— reference / not recordable —	Refe
VI-S3-005	VI	\$3	Referenced Documents	Instr.	SOP-0080295: DLIMS Sample Submission for Manufacturing Sciences		— reference / not recordable —	Refe
VI-S3-006	VI	\$3	Referenced Documents	Instr.	SOP-0080297: Benchtop pH and Conductivity Meter Operation		— reference / not recordable —	Refe
VI-S3-007	VI	\$3	Referenced Documents	Instr.	SOP-0080303: WFI Hold Tank Operation (TK-1458)		— reference / not recordable —	Refe
VI-S3-008	VI	\$3	Referenced Documents	Instr.	SOP-0080390: Operation, Maintenance and Cleaning of the QuattroFlow 4400S, 4000S and 1200S Pumps		— reference / not recordable —	Refe
VI-S3-009	VI	\$3	Referenced Documents	Instr.	SOP-0080402: Operation of the GE Bioprocess Chromatography System in the Gaithersburg Manufacturing Facility		— reference / not recordable —	Refe
VI-S3-010	VI	\$3	Referenced Documents	Instr.	SOP-0080423: Operation and Maintenance of the LevTech and LevMixer Mixing Systems		— reference / not recordable —	Refe
VI-S3-011	VI	\$3	Referenced Documents	Instr.	SOP-0080438: Operation and Cleaning of the Endress and Hauser Promass 80 Flow Meter		— reference / not recordable —	Refe
VI-S3-012	VI	\$3	Referenced Documents	Instr.	SOP-0080503: BriskHeat Full-Coverage Drum Heater Set-up, Operation, and Cleaning		— reference / not recordable —	Refe
VI-S3-013	VI	\$3	Referenced Documents	Instr.	SOP-0080613: Operation of the Millistak+ HC Pod System		— reference / not recordable —	Refe
VI-S3-014	VI	\$3	Referenced Documents	Instr.	SOP-0080839: Peristaltic Pump Operation		— reference / not recordable —	Refe

VI-S3-015	VI	\$3	Referenced Documents	Instr.	SOP-0080999: Handling and Exterior Cleaning of Pressure Gauges		— reference / not recordable —	Refe
VI-S3-016	VI	\$3	Referenced Documents	Instr.	SOP-0081015: Operation of the MagMix Bag System for Low pH Viral Inactivation		— reference / not recordable —	Refe
VI-S3-017	VI	\$3	Referenced Documents	Instr.	SOP-0107056: cGMP Labeling Practices for Controlled Materials, Equipment and Product		— reference / not recordable —	Refe
VI-S3-018	VI	\$3	Referenced Documents	Instr.	SOP-0107091: Operation and Maintenance of the SoloVPE System for Protein Concentration Determination		— reference / not recordable —	Refe
VI-S3-019	VI	\$3	Referenced Documents	Instr.	SOP-0107095: Operation of the Product Hold Fixed Tanks in the Gaithersburg Pilot Facility		— reference / not recordable —	Refe
VI-S3-020	VI	\$3	Referenced Documents	Instr.	SOP-0107097: Sampling from Process Vessels		— reference / not recordable —	Refe
VI-S3-021	VI	\$3	Referenced Documents	Instr.	SOP-0107099: Transfer Line Configuration for Purification		— reference / not recordable —	Refe
VI-S3-022	VI	\$3	Referenced Documents	Instr.	SOP-0107102: Operation and Maintenance of the Integrity Magnetic Mixer Single-Use System		— reference / not recordable —	Refe
VI-S3-023	VI	\$3	Referenced Documents	Instr.	SOP-0107117: Thermometer Use		— reference / not recordable —	Refe
VI-S3-024	VI	\$3	Referenced Documents	Instr.	SOP-0110853: Gowning And Personnel Flow for the Gaithersburg Pilot Facility		— reference / not recordable —	Refe
VI-S3-025	VI	\$3	Referenced Documents	Instr.	SOP-0081641: Manufacturing Sciences SAP Usage at Gaithersburg Pilot Facilities		— reference / not recordable —	Refe
VI-S3-026	VI	\$3	Referenced Documents	Instr.	MABR-0027570 PN: 8012440 / AZD0543 - Affinity Chromatography (DurA Cycle A50, GPF, 2000 L Scale, Process 1)		— reference / not recordable —	Refe
VI-S3-027	VI	\$3	Referenced Documents	Instr.	MABR-0027573 PN: 8012472 / AZD0543 - Anion Exchange Chromatography (CaptoQ, GPF, 2000 L Scale, Process 1)		— reference / not recordable —	Refe
VI-S4-001	VI	\$4	Bill of Materials – Equipment and Components	Fields		Part No./ Equipment ID Quantity Chromatography Skid Mixer Drive Peristaltic Pump Quattroflow Pump pH / Conductivity meter Thermometer Floor Scale / Benchtop Balance Flow Meter Dip tube Clean Hosing 200 L Portable Tank 200 L Portable Tank Fixed Tanks 1500 L Product Hold Tank 2000 L Product Hold Tank 2000 L Product Hold Tank 2500 L Product Hold Tank 4000 L Product Hold Tank Product Filters for Portable/Fixed Product Tanks Filter – 10” SHC Cartridge 0.2 µm Hydrophilic Filter – 5” SHC Cartridge, 0.2 µm Hydrophilic Filter: Fluorodyne EX Grade EDF Cartridges (10 in.) Vent Filters for Portable/Fixed Product Tanks Filter – Aervent 0.2 µm, Hydrophobic, PTFE Membrane (5 in) Filter – Aervent 0.2	SMPL: Display BOM	Cove

						µm, Hydrophobic, PTFE Membrane (10 in)		
VI-S4-002	VI	\$4	Bill of Materials – Equipment and Components	Instr.	4 Bill of Materials – Equipment and Components (continued)	Part No./ Equipment ID Quantity Product Filters for Bags Filter – SHC 0.5+0.2 µm Capsules with C-Flex (3”) Filter – SHC 0.5+0.2 µm Capsules with C-Flex (10”) Filter – 30 in., Opticap® XLT 30 Capsule Filter with Millipore Express® SHC Hydrophilic Membrane, 0.5/0.2 µm Filter – 10 in., Opticap® XLT 10 Capsule filter with Millipore Express® SHC membrane, 0.5/0.2 µm Mixing Bags for Product Filtration Only LevTech Mix Bag (50 L) LevTech Mix Bag (100 L) LevTech Mix Bag (200 L) LevTech Mix Bag (300 L) LevTech Mix Bag (560 L) Mixing Bags for Product Filtration and Viral Inactivation Treatment Custom MagMix Bag (200 L) Custom MagMix Bag (300 L) Custom MagMix Bag (370 L) Custom MagMix Bag (560 L) Bags for WFI Bags - Stedim 3D Bags with C-Flex (500 L) Bags - Stedim 3D Bags with C-Flex (300 L) Bags - Stedim 3D Bags with C-Flex (200 L)	SMPL: Display BOM	Cove
VI-S4-003	VI	\$4	Bill of Materials – Equipment and Components	Instr.	* = Materials with bolded filter part numbers require autoclaving prior to use.		SMPL: Display BOM	Cove
VI-S4-004	VI	\$4	Bill of Materials – Equipment and Components	Instr.	Additional Manufacturing Supplies Used in Process		SMPL: Display BOM	Cove
VI-S4-005	VI	\$4	Bill of Materials – Equipment and Components	Note	If additional materials (filter, bag, etc.) need to be used during processing:		SMPL: Display BOM	Cove
VI-S4-006	VI	\$4	Bill of Materials – Equipment and Components	Instr.	Prior to use, ensure the material is acceptable for use based on Section 4.		SMPL: Display BOM	Cove
VI-S4-007	VI	\$4	Bill of Materials – Equipment and Components	Instr.	Record the applicable material information.		SMPL: Display BOM	Cove
VI-S4-008	VI	\$4	Bill of Materials – Equipment and Components	Instr.	Document the reason for the change in Attachment 7: Comments Section.		SMPL: Display BOM	Cove
VI-S5-001	VI	\$5	Document Process Flow Sheet	Fields		Starting Material PN: 8012440: AZD0543 Affinity Product Acid Addition PN: A-005168: 1M Acetic Acid VI Incubation (Target 75 minutes; NOR 60-240 minutes) Base Addition PN:	SMPL: Additional Solution Batches (Solution Switch)	Cove

						A0101-D: 1M Tris Base Recorded by / date		
VI-S5-002	VI	\$5	Document Process Flow Sheet	Instr.	Additional Solution Batches Used in Process		SMPL: Additional Solution Batches (Solution Switch)	Cove
VI-S5-003	VI	\$5	Document Process Flow Sheet	Note	If a solution (of a different batch noted in Section 5) needs to be replaced during processing:		SMPL: Additional Solution Batches (Solution Switch)	Cove
VI-S5-004	VI	\$5	Document Process Flow Sheet	Instr.	Indicate the switch at the step it is occurring. Ensure the new solution is the same part number as the solution being replaced.		SMPL: Additional Solution Batches (Solution Switch)	Cove
VI-S5-005	VI	\$5	Document Process Flow Sheet	Instr.	Record the solution part number, batch number (and/or process order number) and expiration date below.		SMPL: Additional Solution Batches (Solution Switch)	Cove
VI-S5-006	VI	\$5	Document Process Flow Sheet	Instr.	Document the change in Attachment 7: Comments Section.		SMPL: Additional Solution Batches (Solution Switch)	Cove
VI-S6-001	VI	\$6	Process Notes	Note	All activities are performed at 15 °C to 27 °C.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-002	VI	\$6	Process Notes	Note	For all calculations, assume 1 L = 1 kg and 1 mL = 1 g unless otherwise noted.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-003	VI	\$6	Process Notes	Note	Verify that all solutions, raw materials, and standards are within expiration prior to use.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-004	VI	\$6	Process Notes	Note	When an equipment ID is written in the MPR, the operator is indicating they have visually inspected and verified calibration prior to use.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-005	VI	\$6	Process Notes	Note	If additional mixing/recirculation is required after initial sampling of the product, record the mixing information in Attachment 6: Product Recirculation Worksheet.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-006	VI	\$6	Process Notes	Note	If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-007	VI	\$6	Process Notes	Note	If Non-Routine samples are taken, re-adjust the weight of the product sampled.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S6-008	VI	\$6	Process Notes	Note	All parameters listed in this document are Targets/Alert Limits unless otherwise noted. When alert limits are exceeded, contact Supervisor and Technical Transfer Group. Document any actions taken in the Attachment 7: Comments Section of the MPR.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
VI-S7-001	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Record the room where this process will occur.	Room Number	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-002	VI	\$7	Affinity Product Information / Pre-Processing Activities	Note	If the Affinity Product was warmed up (from cold storage), ensure the time the product was removed from cold storage is recorded in Attachment 3: Hold Times.		SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-003	VI	\$7	Affinity Product Information / Pre-	Instr.	Confirm that the pH meter has been standardized between pH 2.0 and pH 7.0 per SOP-0080297. Confirm	Meter ID	SMPL: Incoming Product Information (Weight /	Cove

			Processing Activities		that the conductivity meter has been standardized within the acceptable range per SOP-0080297.		Concentration / DLIMS)	
VI-S7-004	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Confirm that the thermometer is within its calibration interval. Record the equipment ID of the thermometer used.	Thermometer ID	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-005	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Record the tare weight and net weight of the AZD0543 – Affinity Product vessel.	Tare Weight of the Affinity Product Vessel kg = L Net Weight of the Affinity Product kg = L Affinity Product SoloVPE Concentration (8012440-30) DLIMS Project Number DLIMS Sample Number	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-006	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Mix the product for at least 15 minutes. Follow SOP-0080423 for the LevTech mixer, if applicable. Follow SOP-0107102 for the Magnetic Mixer, if applicable. Follow SOP-0107095 for fixed tanks, if applicable. NOTE: If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.	Mixer ID/Tank ID Agitation Setting Mixing Start Time Mixing End Time Mixing Duration (≥ 15 minutes)	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-007	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Remove at least the sample volume per Attachment 1: Sampling Record and SOP-0107097. If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.	Sampling Date and Time Date: Time:	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-008	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Aliquot the sample (8012441-10) per Attachment 1: Sampling Record. If applicable, aliquot non-routine samples per Attachment 2: Non-Routine Sampling Record. Label the sample per SOP-0107056.		SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-009	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Submit the sample to the storage location indicated in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.		SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-010	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Measure the pH, conductivity, and temperature of the Affinity product per SOP-0080297 and SOP-0107117. Record the results below.	Conductivity mS/cm Temperature	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-011	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Obtain the gross weight of the Affinity product vessel after sampling. NOTE: Ensure the vessel is in the same conformation as when the tare weight was obtained.	Scale/Balance ID Calibration Due Date Gross Weight/Volume of the Affinity Product vessel after sampling kg=L	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-012	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Calculate the net weight of the Affinity product after sampling:	Gross Weight/Volume of the Affinity Product vessel after sampling (Step 7.11) Tare Weight of the Affinity Product Vessel (Step 7.5) Affinity Weight/Volume after sampling kg=L	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-013	VI	\$7	Affinity Product Information / Pre-Processing Activities	Note	If the product will be transferred to a different adjustment vessel, N/A Step 7.14 and go to Section 8. Otherwise: N/A Section 8 and re-label the Viral Inactivation (VI) vessel “VI Treatment Vessel” and record the tare weight on the label. Continue to Step 7.14.	Starting Product Temperature Check Proceed per the table below. Check One Temperature of the sample at Step 7.10 is: Action Within 18 – 25 oC	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-014	VI	\$7	Affinity Product Information / Pre-	Instr.	Transcribe the pH, conductivity, and temperature results into Step 10.9 “Initial pH / Conductivity / Temperature.”	Less than 18 oC:	SMPL: Incoming Product Information (Weight /	Cove

			Processing Activities		N/A all of Section 9 and continue to Step 10.1.		Concentration / DLIMS)	
VI-S7-015	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Turn on the treatment vessel's agitator and document the mixing start time in Step 9.1. * Begin warming the vessel per SOP-0080503/SOP-0080023 (for a bag) or SOP-0107095 (for a tank) to 18 – 25 oC. ** Continue to Section 9.	Greater than 25 oC	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-016	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	Contact a supervisor and document actions taken in Attachment 7: Comments Section.		SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S7-017	VI	\$7	Affinity Product Information / Pre-Processing Activities	Instr.	* NOTE: The product warming range is 18 – 25 oC. ** NOTE: If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.		SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	Cove
VI-S8-001	VI	\$8	Transfer Vessel Information and Set up	Instr.	NOTE: If the Affinity product vessel will be used as the VI Treatment Vessel, N/A all of Section 8, then continue to Section 9 or Section 10, depending on the Action selected in Step 7.14.		SMPL: VI Treatment Vessel Setup	Cove
VI-S8-002	VI	\$8	Transfer Vessel Information and Set up	Instr.	Obtain the product vessel and label the vessel with "AZD0543", "VI Treatment Vessel", batch number, part number, initials, and date.		SMPL: VI Treatment Vessel Setup	Cove
VI-S8-003	VI	\$8	Transfer Vessel Information and Set up	Instr.	For Step 8.4: VI Treatment Vessel Information: Obtain the tare weight of the Treatment Vessel and record the the information in Step 8.4.	VI Treatment Vessel Bag Information (if applicable) Bag Information Mark the applicable attachments connected to the mixer bag during the vessel tare weight Attached: Mark the applicable attachments connected to the mixer bag during the vessel tare weight Mixer Drive Top Base Magnetic Clamp Tank Information (if applicable) Tank ID CIP Batch ID CIP Exp. Date / Time Date: Time: Pressure Test ID Pressure Test Date SIP Batch ID Filter Information (See Section 4 for acceptable filters and autoclaving requirements.) Vent Filter (if applicable) Product Filter 1 Product Filter 2 Product Filter 3 Tare Weight Information * (NOTE: If collection vessel is a bag, tare weight will be obtained prior to attachment of the product filter.) Scale/ Balance ID Calibration Due Date Tare Weight of the VI Treatment Vessel * Mark filters attached when tare weight is obtained. Product filter Vent filter No filters, vessel is a bag Recorded by / Date	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-004	VI	\$8	Transfer Vessel Information and Set up	Instr.	Obtain transfer tubing or a clean flex hose to transfer product from the Affinity Product Vessel to the VI		SMPL: VI Treatment Vessel Setup	Cove

					Treatment Vessel. Record the tubing or hose information in Attachment 5: Transfer Tubing/Hosing Information.			
VI-S8-005	VI	\$8	Transfer Vessel Information and Set up	Instr.	Connect the transfer tubing/hose to the outlet of the Affinity Product Vessel and to the inlet of the VI Treatment Vessel. Thread the transfer tubing through a peristaltic pump, if applicable. Begin transferring the Affinity product to the VI Treatment Vessel.		SMPL: VI Treatment Vessel Setup	Cove
VI-S8-006	VI	\$8	Transfer Vessel Information and Set up	Instr.	After all of the Affinity product has been transferred to the VI Treatment Vessel, disconnect the transfer line from the VI Treatment Vessel.		SMPL: VI Treatment Vessel Setup	Cove
VI-S8-007	VI	\$8	Transfer Vessel Information and Set up	Instr.	Mix the product for at least 15 minutes. Follow SOP-0107102 for the MagMix, if applicable. Follow SOP-0107095 for fixed tanks, if applicable. NOTE: If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.	Mixer ID/Tank ID Agitation Setting Mixing Start Time Mixing End Time Mixing Duration (≥15 minutes)	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-008	VI	\$8	Transfer Vessel Information and Set up	Instr.	Aliquot a 10 mL product sample from the treatment vessel into a 50 mL conical and measure the sample temperature. Record the results below.	Temperature	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-009	VI	\$8	Transfer Vessel Information and Set up	Instr.	Obtain the gross and net weight of the Affinity product after sampling.	Scale/Balance ID Calibration Due Date Gross Weight/Volume of the Affinity Product after sampling Net Weight/Volume of the Affinity Product after sampling * * Net Weight/Volume of the Affinity Product after sampling = Gross Weight/Volume of the Affinity Product after sampling (above) – Tare weight of the Product Vessel (Step 8.4) Starting Product Temperature Check Proceed per the table below. Check One Temperature of the sample at Step 8.9 is: Action Within 18 – 25 oC	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-010	VI	\$8	Transfer Vessel Information and Set up	Instr.	Transcribe the pH, conductivity, and temperature results into Step 10.9 "Initial pH / Conductivity / Temperature." N/A all of Section 9 and continue to Section 10.	Less than 18 oC:	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-011	VI	\$8	Transfer Vessel Information and Set up	Instr.	Turn on the treatment vessel's agitator and document the mixing start time in Step 9.1. * Begin warming the vessel per SOP-0080503/SOP-0080023 (for a bag) or SOP-0107095 (for a tank) to reach 18 – 25oC. ** Continue to Section 9.	Greater than 25 oC	SMPL: VI Treatment Vessel Setup	Cove
VI-S8-012	VI	\$8	Transfer Vessel Information and Set up	Instr.	Contact a supervisor and document actions taken in Attachment 7: Comments Section.		SMPL: VI Treatment Vessel Setup	Cove
VI-S8-013	VI	\$8	Transfer Vessel Information and Set up	Instr.	* NOTE: The product warming range is 18 – 25 oC. ** NOTE: If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.		SMPL: VI Treatment Vessel Setup	Cove

VI-S9-001	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Start mixing the product per SOP-0107102 (for a bag) or SOP-0107095 (for a tank) and record the mixing start time. If the product requires recirculation to mix, N/A rows below and complete Attachment 6: Product Recirculation Worksheet.	Mixer ID/Tank ID Agitation Setting Mixing Start Time	SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-002	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Aliquot a 10 mL product sample from the treatment vessel into a 50 mL conical and measure the sample temperature. Record the results below.	Temperature Product Temperature Check Proceed based on the table below. Check One Temperature of the sample at Step 9.2 is: Action Within 18 – 25 oC	SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-003	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 "Initial pH / Conductivity / Temperature." Document the mixing end time in Step 9.4, then continue to Section 10.	Less than 18 oC	SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-004	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Continue to warm and agitate the product. * Resample the product regularly until the product is within 18 – 25 oC. When the product is within 18 – 25 oC, measure the pH, conductivity, and temperature of the sample and transcribe results into Step 10.9 "Initial pH / Conductivity / Temperature." Document the mixing end time in Step 9.4, then continue to Section 10.	Greater than 25 oC	SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-005	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Contact a supervisor and document actions taken in Attachment 7: Comments Section.		SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-006	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	* If using a tank, activating the glycol loop may alter the tank weight. For this reason, the glycol loop must remain on through neutralization to minimize weight fluctuations during acid and base additions.		SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S9-007	VI	\$9	Additional Mixing and Pre-Treatment Product Warming (if applicable)	Instr.	Record the mixing end time. Calculate the mixing duration from the time the mixing was started in Step 9.1.	Mixing End Time Mixing Duration (≥15 minutes)	SMPL: Starting / Transfer Temperature Decision (block stack)	Cove
VI-S10-001	VI	\$10	Setup and Acid Addition	Instr.	As needed, obtain tubing and/or hosing for the transfer of the acid to the starting material. Record the tubing / hosing information in Attachment 5: Transfer Tubing/Hosing Information.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-002	VI	\$10	Setup and Acid Addition	Note	If using a tank, confirm that the VI Treatment Vessel has a dip tube on the inlet and that the dip tube outlet is below the surface of the product in the treatment vessel.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-003	VI	\$10	Setup and Acid Addition	Instr.	Attach tubing/hosing from the outlet of the acid buffer vessel (A-005168) to the inlet of the VI Treatment Vessel, as applicable. Follow SOP-0081015 for a MagMix bag. If applicable, thread the tubing through a peristaltic pump following SOP-0080839. Prime the tubing/hosing leading to the VI Treatment Vessel with (A-005168) pump per SOP-0080839.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove

VI-S10-004	VI	\$10	Setup and Acid Addition	Instr.	Record the gross weight of the VI Treatment Vessel after priming and the scale/balance used in Step 10.9.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-005	VI	\$10	Setup and Acid Addition	Instr.	Ensure agitation is on.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-006	VI	\$10	Setup and Acid Addition	Instr.	Calculate the expected amount of acid to be added based on the addition ratio. * Use Step 8.10 if the product was transferred.	Net Weight of Affinity after sampling (Step 7.12 or 8.10 *) Acid Addition Ratio (kg acid/ kg Affinity Product) Recommended weight of acid	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-007	VI	\$10	Setup and Acid Addition	Instr.	Determine the initial acid addition weight. After initial addition, more acid may be added until the target pH is reached.	Recommended weight of acid (Step 10.6) Initial Addition Ratio Target Weight for Initial Acid Addition	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-008	VI	\$10	Setup and Acid Addition	Instr.	Step 10.9: Acid Treatment Begin transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 10.9.	Acid Treatment Initial Measurements Initial pH Initial Conductivity Initial Temperature Start time of addition (End of storage time) Initial Measurements mS/cm Scale/Balance ID Calibration due date Gross weight of Treatment Vessel before addition (kg) End time of acid addition Gross weight of Treatment Vessel after addition (kg) Net weight of buffer added (kg) a Total net weight of buffer added (kg) b Time of pH sampling Adjusted pH Sample Temperature (oC)	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-009	VI	\$10	Setup and Acid Addition	Instr.	Note: Do not exceed the volume calculated in Step 10.7 on the initial addition.	Time of pH sampling Adjusted pH Sample Temperature (oC) 10.9 Acid Treatment Cont. Time of pH sample confirmed at 3.5 c, d Final pH (3.4 – 3.6) c Final Conductivity mS/cm Temperature (18 – 25 oC) e Start Time for Incubation (pH and temperature in-spec) f	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-010	VI	\$10	Setup and Acid Addition	Instr.	Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-011	VI	\$10	Setup and Acid Addition	Instr.	Total net weight of buffer added = Net weight of buffer added (kg) + Total net weight of buffer added (kg) from the previous addition. For the first addition only—the previous net weight is zero.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-012	VI	\$10	Setup and Acid Addition	Instr.	The target pH is 3.5. The acid addition may be continued after the pH is within range to be closer to the target pH.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-013	VI	\$10	Setup and Acid Addition	Instr.	This time should match the time of the first sample where pH is confirmed within 3.4 – 3.6.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S10-014	VI	\$10	Setup and Acid Addition	Instr.	The NOR temperature range for the duration of incubation is 18 – 25 oC.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove

VI-S10-015	VI	\$10	Setup and Acid Addition	Instr.	Start Time for Incubation (pH and temperature in spec) should match the time of the first sample when the pH is confirmed to be within 3.4 – 3.6 and temperature 18 – 25 oC.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S11-001	VI	\$11	Incubation	Instr.	Mid-Incubation Temperature Check for a Bag Vessel If the treatment vessel is a tank, N/A this step. Otherwise, continue below. Incubate the product for a minimum of 38 minutes from Step 10.9 "Start Time for Incubation (pH and temperature in spec)," then sample 10 mL from the treatment vessel. Measure the sample temperature. If the sample is outside the range of 18 – 25 oC, contact a supervisor and document actions taken in Attachment 7: Comments Section.	Sampling Time Temperature	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S11-002	VI	\$11	Incubation	Instr.	Incubated Product Information Confirm that the product has been incubating for a minimum of 60 minutes. Take a 25 mL sample from the incubated product. Measure the pH, conductivity, and temperature of the sample and record the results below.	Time of Sampling pH (3.4 – 3.6) Conductivity mS/cm Temperature (18 – 25 oC)	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S11-003	VI	\$11	Incubation	Instr.	Calculate the duration that the product has been incubating.	Time of Sampling (Step 11.2) Start Time for Incubation (pH and temperature in spec) (Step 10.9) Duration the product has been incubating	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S12-001	VI	\$12	Acid Addition- Extension	Instr.	Choose an option based on the Step 11.2 sample result:	Check One Condition Action The pH is above 3.6 Continue to the next step. The pH is within 3.4 – 3.6 and temperature is within 18 – 25 oC N/A the remainder of this section and Section 13 Go to Section 14. The pH is below 3.4 or the temperature is not within 18 – 25 oC	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-002	VI	\$12	Acid Addition- Extension	Instr.	Contact an area manager and document actions taken in Attachment 7: Comments Section.	Name of Supervisor Contacted, if applicable	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-003	VI	\$12	Acid Addition- Extension	Instr.	Ensure agitation is on.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-004	VI	\$12	Acid Addition- Extension	Instr.	Step 12.4: Acid Treatment Extension Continue transferring 1M Acetic Acid (A-005168) into the VI Treatment Vessel: Do not stop agitation during Treatment. Follow SOP-0081015 for performing VI in a MagMix bag. If the VI Treatment Vessel is a tank, follow SOP-0107095. After each addition of A-005168, mix for at least ten minutes before sampling for pH measurement. Continue adding A-005168 until the product pool is within the following ranges: pH Range (NOR): 3.4 - 3.6 (Target pH is 3.5) Temperature Range: 18 – 25 oC If necessary, warm the product per SOP-0080503/SOP-0080023 (if using a bag) or SOP-0107095 (if using a tank). Record the Final pH, Final conductivity, Temperature, and Start Time for Incubation (pH and temperature in spec) in Step 12.4.	Acid Treatment - Extension Scale/Balance ID Calibration Due Date Start time of addition Gross weight of Treatment Vessel before addition (kg) End time of acid addition Gross weight of Treatment Vessel after addition (kg) Net weight of buffer added (kg) a Total net weight of buffer added (kg) b Time of pH sampling Adjusted pH Sample Temperature (oC) Time of pH sample confirmed at 3.5 c, d Final pH (3.4 – 3.6) c Final Conductivity ----- mS/cm Temperature (18 – 25 oC) e Start Time for Incubation (pH and Temperature in spec) f	SMPL: Iterative pH Titration Table (Acid / Base)	Cove

VI-S12-005	VI	\$12	Acid Addition- Extension	Instr.	Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-006	VI	\$12	Acid Addition- Extension	Instr.	Total net weight of buffer added = Net weight of buffer added (kg) +Total net weight of buffer added (kg) from the previous addition. For the first addition only—use the total amount of buffer added in Step 10.9 as the “Total Net Weight of Buffer added.”		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-007	VI	\$12	Acid Addition- Extension	Instr.	The target pH is 3.5. The acid addition may be continued after the pH is within range to be closer to the target pH.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-008	VI	\$12	Acid Addition- Extension	Instr.	This time should match the time of the first sample where pH is confirmed within 3.4 – 3.6.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-009	VI	\$12	Acid Addition- Extension	Instr.	The NOR temperature range for the duration of incubation is 18 – 25 oC.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S12-010	VI	\$12	Acid Addition- Extension	Instr.	Start Time for Incubation (pH and Temperature in spec) should match the time of the first sample that pH is confirmed to be within 3.4 – 3.6 and temperature 18 – 25 oC.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S13-001	VI	\$13	Incubation - Extension	Instr.	Mid-Incubation Temperature Check for a Bag Vessel: If the treatment vessel is a tank, N/A this step. Otherwise, continue below. Incubate the product for a minimum of 38 minutes from Step 12.4 “Start Time for Incubation (pH and Temperature in spec),” then sample 10 mL from the treatment vessel. Measure the sample temperature. If the sample is outside the range of 18 – 25 oC, contact a supervisor and document actions in Attachment 7: Comments Section.	Sampling Time Temperature	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S13-002	VI	\$13	Incubation - Extension	Instr.	Incubated Product Information Confirm that the product has been incubating for a minimum of 60 minutes. Take a 25 mL sample from the incubated product. Measure the pH, conductivity, and temperature of the sample and record the results below.	Time of Sampling pH (3.4 – 3.6) Conductivity mS/cm Temperature (18 – 25 oC)	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S13-003	VI	\$13	Incubation - Extension	Instr.	Calculate the duration that the product has been incubating.	Time of Sampling (Step 13.2) Start Time for Incubation (pH and Temperature in spec) (Step 12.4) Duration that the product has been incubating	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S13-004	VI	\$13	Incubation - Extension	Instr.	Choose an option based on the Step 13.2 sample result.	Check One Condition Action The pH is within 3.4 – 3.6 and the temperature is within 18 – 25 oC. Continue to Section 14	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S13-005	VI	\$13	Incubation - Extension	Instr.	The pH is not within 3.4 – 3.6 or the temperature is not within 18 – 25 oC.		SMPL: Incubation Timing & Incubated Sample (block stack)	Cove
VI-S13-006	VI	\$13	Incubation - Extension	Instr.	Contact an area manager and document actions taken in Attachment 7: Comments Section.	Name of Supervisor Contacted, if applicable	SMPL: Incubation Timing & Incubated Sample (block stack)	Cove

VI-S14-001	VI	\$14	Setup and Base Addition	Instr.	Disconnect the acid transfer tubing from the VI Treatment vessel. Drain the tubing used for acid addition.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-002	VI	\$14	Setup and Base Addition	Note	If using a tank and the glycol loop was activated to adjust the treatment vessel temperature, keep the glycol loop on during neutralization to prevent fluctuations to the vessel weight during base addition. If using a bag and a heating blanket was used to adjust the vessel temperature, remove the blanket before proceeding with base addition. If the product did not need to be temperature adjusted, N/A this step and continue to Step 14.3.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-003	VI	\$14	Setup and Base Addition	Instr.	Attach tubing/hosing from the outlet of the base buffer vessel 1M Tris Base (A0101-D) to the inlet of the VI Treatment Vessel, as applicable. If using a MagMix bag, follow SOP-0081015. If applicable, thread the tubing through a peristaltic pump following SOP-0080839. Prime the tubing/hosing leading to the VI Treatment Vessel with (A0101-D).		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-004	VI	\$14	Setup and Base Addition	Instr.	Record the gross weight of the VI Treatment Vessel after priming in Step 14.11.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-005	VI	\$14	Setup and Base Addition	Instr.	Ensure agitation is on.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-006	VI	\$14	Setup and Base Addition	Instr.	Calculate the expected amount of base to be added based on the addition ratio.	Total Net Weight of Acid added (Step 10.9 or 12.4 *) Base Addition Ratio (kg base / kg acid) Recommended weight of base	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-007	VI	\$14	Setup and Base Addition	Instr.	* If additional acid addition was needed, use Step 12.4. Ensure the acid from the first addition is included in the total net weight of acid added.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-008	VI	\$14	Setup and Base Addition	Instr.	Determine the initial base addition weight. After initial addition, more base may be added to reach the target pH.	Recommended weight of base (Step 14.6) Initial Addition Ratio Target Weight of Initial Base Addition	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-009	VI	\$14	Setup and Base Addition	Instr.	Steps 14.9 and 14.10 provide instructions for filling out the table in Step 14.11 and performing the base additions. Review the instructions in Steps 14.9 and 14.10 before proceeding.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-010	VI	\$14	Setup and Base Addition	Instr.	For Step 14.11 - End of Incubation Information: Take a final sample of the incubation product and measure the pH and temperature. In Step 14.11: Record the (A) 'pH at the end time of Incubation' and (B) 'Temperature at the end of Incubation' Record the (C) 'End time for Incubation', which is also the start time of the first base addition. Calculate the (D) 'Total Duration of Incubation' NOTE: Contact a supervisor if the Total Duration of Incubation exceeds 240 minutes and document actions taken in Attachment 7: Comments Section.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove

VI-S14-011	VI	\$14	Setup and Base Addition	Instr.	For Step Step 14.11 - Neutralization Treatment: After a target incubation time of 75 minutes, within a range of 60 – 240 minutes, begin transferring 1M Tris Base (A0101-D) into the VI Treatment Vessel: Do not stop agitation during treatment. For the base additions, follow SOP-0081015 if the VI Treatment Vessel is a MagMix bag. If the vessel is a tank, follow SOP-0107095. After each addition of A0101-D, mix the product for at least ten minutes prior to taking a sample for pH measurement. When pH rises above 4.0, recalibrate the pH meter to pH 4.0 and pH 10.0 and record the calibration information in Step 14.12. Continue adding A0101-D until the product pool is within pH range 7.2 – 7.6. Record the Final pH, Final conductivity, Temperature and End Time For Incubation in Step 14.11.	Neutralization Information End of Incubation Information (A) pH at the end time of Incubation (3.4 – 3.6) (B) Temperature at the end of Incubation (18 – 25 oC) (C) End time for Incubation (Start Time of base addition) Start Time of Incubation (Step 10.9 or 12.4) a (D) Total Duration of Incubation (60-240 minutes) minutes Neutralization Treatment Scale/Balance ID Calibration Due Date Gross weight of Treatment Vessel before addition (kg) End time of base addition Gross weight of Treatment Vessel after addition (kg) Net weight of buffer added (kg) b Total net weight of buffer added (kg) c Time of pH sampling d Adjusted pH e	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-012	VI	\$14	Setup and Base Addition	Instr.	Note: Do not exceed the volume calculated in Step 14.7 on the initial addition.	Time of pH sampling d Adjusted pH e Final pH (7.2 – 7.6) e Final Conductivity mS/cm Temperature (15 – 27 °C)	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-013	VI	\$14	Setup and Base Addition	Instr.	Use Step 12.4 if the acid addition was extended.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-014	VI	\$14	Setup and Base Addition	Instr.	Net weight of buffer added (kg) = Gross weight of Treatment Vessel after addition (kg) – Gross weight of Treatment Vessel before addition (kg).		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-015	VI	\$14	Setup and Base Addition	Instr.	Total net weight of buffer added = Net weight of buffer added (kg) +Total net weight of buffer added (kg) from the previous addition. For the first addition only—the previous net weight is zero.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-016	VI	\$14	Setup and Base Addition	Instr.	Re-standardization of the benchtop pH meter is needed at Step 14.12 when the pH is $\geq$ 4.0.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-017	VI	\$14	Setup and Base Addition	Instr.	The target pH after neutralization is 7.4. The NOR pH after neutralization is 7.2 – 7.6.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-018	VI	\$14	Setup and Base Addition	Instr.	Record the standardization of the benchtop pH meter with pH 4.0 and pH 10.0 standards.	pH Meter ID	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-019	VI	\$14	Setup and Base Addition	Instr.	Once the product pH has been confirmed to be in neutralization range: Close the outlet of the base vessel and the inlet of the VI Treatment Vessel. Disconnect the base transfer tubing from the VI Treatment Vessel. Drain the tubing used for base addition.		SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S14-020	VI	\$14	Setup and Base Addition	Instr.	Obtain and record the gross weight of the VI Treatment Vessel after the neutralization. NOTE: Ensure the VI Treatment Vessel is in the same configuration as when the tare weight was obtained in Step 7.5 (or Step 8.4 if product was transferred to a treatment vessel).	Scale/Balance ID Calibration Due Date Gross Weight after Neutralization	SMPL: Iterative pH Titration Table (Acid / Base)	Cove

VI-S14-021	VI	\$14	Setup and Base Addition	Instr.	Calculate the net weight of the Neutralized VI product. * Use Step 8.10 if the Affinity Product was transferred to a treatment vessel. ** Use Step 12.4 if additional acid was needed. Ensure that the acid from the first addition is included in the total net weight of acid added.	Net Weight of Affinity Product after sampling (Step 7.12 or 8.10 *) Total Net Weight of Acid added (Step 10.9 or 12.4 **) Total Net Weight of Base added (Step 14.11) Net Weight/Volume of the Neutralized VI Product kg=L	SMPL: Iterative pH Titration Table (Acid / Base)	Cove
VI-S15-001	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Mix the product for at least 15 minutes. Follow SOP-0080423 for the LevTech mixer, if applicable. Follow SOP-0107102 for the Magnetic Mixer, if applicable. Follow SOP-0107095 if using a product hold tank, if applicable. NOTE: If the product is in a non-mixing vessel and requires recirculation to mix, N/A the step below and complete Attachment 6: Product Recirculation Worksheet.	Mixer ID/Tank ID Agitation Setting Mixing Start Time Mixing End Time Mixing Duration (≥ 15 minutes)	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-002	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Remove at least the sample volume per Attachment 1: Sampling Record and SOP-0107097. If a non-routine sample is requested by memo, complete Attachment 2: Non-Routine Sampling Record.	Sampling Date and Time Date: Time:	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-003	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Aliquot the sample (8012441-30) per Attachment 1: Sampling Record. If applicable, aliquot non-routine samples per Attachment 2: Non-Routine Sampling Record. Label the sample per SOP-0107056.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-004	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Submit the sample to the storage location indicated in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-005	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Measure the pH, conductivity, and temperature of the sample. NOTE: Contact Supervisor if the pH of the VI Product is not 7.2 – 7.6 and document actions taken in Attachment 7: Comments Section.	pH (7.2 – 7.6) Conductivity mS/cm Temperature	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-006	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Obtain the gross and net weights of the collected product after sampling and record the values.	Scale/Balance ID Calibration Due Date Gross Weight of the VI Treatment Vessel after sampling	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-007	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Calculate the net weight/volume of the VI Product after sampling. * NOTE: Use Step 8.4 as tare weight if the Affinity product was transferred to a treatment vessel in Section 8. Otherwise, use Step 7.5 as tare weight.	Gross Weight of the VI Treatment Vessel after sampling (Step 15.6) Tare Weight of the VI Product Vessel (Step 7.5 or 8.4)* Net Weight/Volume of the VI Product after sampling kg = L	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-008	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Store the product and document storage information below. For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage. For product stored at 15 – 25°C, the storage start time is the time when final sampling occurred in Step 15.2.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S15-009	VI	\$15	Viral-Inactivated Product – Sampling	Instr.	Ensure that Attachment 3: Hold Times is filled out with all hold times for the Affinity product.	Name in MPR Name in Attachment Step No. Start time of addition Affinity Product Storage End (Time/Date) Step 10.9	SMPL: Product Sampling & DLIMS Submission	Cove

VI-S16-001	VI	\$16	Batch Production Record Review	Fields		Production Initial/Date	SMPL: Instruction & Sign-off	Cove
VI-S16-002	VI	\$16	Batch Production Record Review	Instr.	Ensure any additional supporting documentation is included (e.g. additional Pi Trends, Memos, Logbook pages, Non-Routine sample requests, etc.).		SMPL: Instruction & Sign-off	Cove
VI-S16-003	VI	\$16	Batch Production Record Review	Note	Verify that the product net weight after sampling (from Step 15.7) is correctly inputted into SAP.		SMPL: Instruction & Sign-off	Cove
VI-S16-004	VI	\$16	Batch Production Record Review	Instr.	Review the Batch Production Record for completeness and submit for review.		SMPL: Instruction & Sign-off	Cove
VI-S17-001	VI	\$17	Attachment 1: Sampling Record	Fields		Viral Inactivation Dev Samples DLIMS Project # Pre-Load Performed By/Date DLIMS Test Name Volume Location Tube Type Intermediate Storage Temp b DLIMS Storage Temp Sample Number -10 (Affinity Product) BIOBURDEN_IP ENDOTOXIN_IP_GPF1 PET -30 (VI Product) SOLO_A280 a CIEF_NATIV CHO_SYBR EL_CHO_GEN SEC_PUR CGE_N_RED ABT Retain 5 x 14 mL PPS Small Retain 3 x 14 mL	SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-002	VI	\$17	Attachment 1: Sampling Record	Instr.	Test SOLO_A280 sample per SOP-0107091. If sample is not tested immediately, store sample at 2 - 8°C.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-003	VI	\$17	Attachment 1: Sampling Record	Instr.	The intermediate storage temperature applies to both the shipment and the storage of the sample prior to reaching the sample management team.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-004	VI	\$17	Attachment 1: Sampling Record	Instr.	Use GREENFORM Template for Development DLIMS sample submissions.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-005	VI	\$17	Attachment 1: Sampling Record	Instr.	Sample Designation "-10": For sampling of starting material.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-006	VI	\$17	Attachment 1: Sampling Record	Instr.	Sample Designation "-30": For product quality samples.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S17-007	VI	\$17	Attachment 1: Sampling Record	Instr.	Bold (STAT): Bold text indicates STAT sample submission.		SMPL: Product Sampling & DLIMS Submission	Cove
VI-S18-001	VI	\$18	Attachment 2: Non-Routine Sampling Record	Note	If non-routine samples are taken, re-adjust the weight of the product sample.		SMPL: Non-Routine Sampling Record	Cove
VI-S18-002	VI	\$18	Attachment 2: Non-Routine Sampling Record	Instr.	Log into DLIMS per SOP-0080295 as required by the memo. Record DLIMS Project number and Sample number if submitting via DLIMS.	Pre-Load Performed By/Date Process Section No. Contents DLIMS Form DLIMS Project Number Container Type Quantity DLIMS Tests Sample Number Designation Receiving Group Storage Temp (°C) Sample Performed By/Date Sample Witnessed By/Date	SMPL: Non-Routine Sampling Record	Cove
VI-S19-001	VI	\$19	Attachment 3: Hold Times	Instr.	Record the storage start time, end time, and date for the Affinity Product.		SMPL: Hold Time Table	Cove

VI-S19-002	VI	\$19	Attachment 3: Hold Times	Instr.	For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage, and the storage end time is the time when the product is transferred from 2 – 8°C storage. A new entry for storage at R. T. (15 – 25°C) begins upon removal from 2 – 8°C.		SMPL: Hold Time Table	Cove
VI-S19-003	VI	\$19	Attachment 3: Hold Times	Instr.	Calculate the storage duration for each temperature where the product was stored as well as the combined storage duration, if applicable.	Hold Location Temp. Affinity Product Storage Start (Time/Date) Affinity Product Storage End (Time/Date) Hold Times Hold Location Temp. Affinity Product Storage Start (Time/Date) Affinity Product Storage End (Time/Date) R.T. 2-8 °C ____hrs ____mins R.T. 2-8 °C ____hrs ____mins R.T. 2-8 °C ____hrs ____mins Total R. T. Hold Time: If R.T. Total Hold Time > 72 hrs, contact a supervisor Total 2-8 °C Hold Time Total 2-8 °C + RT Hold Time: If Total Combined Hold > 168 hrs, contact a supervisor	SMPL: Hold Time Table	Cove
VI-S20-001	VI	\$20	Attachment 4: Yield Calculations	Fields		Load Mass Pre-Sampling Load Volume Pre-Sampling Load Concentration Load Mass Pre-Sampling ____L_ Step 7.5 ____g/L_ Step 7.5 Load Mass Post-Sampling Load Volume Post-Sampling Load Concentration Load Mass Post-Sampling Load Volume Post-Sampling Load Concentration ____L_ Step 7.12 or Step 8.10 ____g/L_ Step 7.5 * Use Step 8.10 volume if Affinity product was transferred to a new VI Treatment Vessel Final Product Mass Pre-Sampling Final Product Volume Pre-Sampling Final Product Concentration Final Product Mass Pre-Sampling Final Product Volume Pre-Sampling Final Product Concentration ____L_ Step 14.15 Final Product Mass Post-Sampling Final Product Volume Post-Sampling Product Concentration Final Product Mass Post-Sampling Final Product Volume Post-Sampling Product Concentration ____L_ Step 15.7	SMPL: Yield Calculations	Cove
VI-S20-002	VI	\$20	Attachment 4: Yield Calculations	Instr.	Attachment 4: Yield Calculations (continued)	Final Product Mass Pre-Sampling Load Mass Post-Sampling Yield % Yield Yield % Yield Final Product Mass Post-Sampling Load Mass Pre-Sampling Yield % Yield Yield % Yield	SMPL: Yield Calculations	Cove

VI-S21-001	VI	\$21	Attachment 5: Transfer Tubing/Hosing Information	Fields		Tubing/Hose 1 Tubing/Hose 2, if applicable Tubing/Hose 3, if applicable Applicable MPR Step Tubing Part No, if applicable Tubing Batch No, if applicable Tubing Exp. Date, if applicable Tubing Autoclave Cycle No., if applicable Tubing Autoclave Exp. Date, if applicable Hose Serial No., if applicable Hose Cleaning Batch ID, if applicable Hose Cleaning Exp. Date, if applicable Recorded By / Date	SMPL: Transfer Tubing / Hosing Information	Cove
VI-S22-001	VI	\$22	Attachment 6: Product Recirculation Worksheet	Instr.	Obtain a piece of tubing for recirculation of the product. Record the applicable product name and tubing information. Start mixing the product by recirculation. Record the start time. Mix the product by recirculation per SOP-0107097 for at least 40 minutes and record the mixing information.	Section 7 Section 8 Section 9 Section 15 Product Name Tubing Part No. Tubing Batch No. Tubing Exp. Date Tubing Autoclave cycle No. Tubing Autoclave Exp. Date Mixing Start Time Mixing End Time Mixing Duration * (≥ 40 minutes)	SMPL: Product Recirculation Worksheet	Cove
VI-S22-002	VI	\$22	Attachment 6: Product Recirculation Worksheet	Instr.	* The duration of mixing is the time that elapses between the mixing end time and the mixing start time in minutes.		SMPL: Product Recirculation Worksheet	Cove
VI-S23-001	VI	\$23	Attachment 7: Comments Section	Fields		Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
VI-S23-002	VI	\$23	Attachment 7: Comments Section	Instr.	Attachment 7: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
VI-S23-003	VI	\$23	Attachment 7: Comments Section	Instr.	Attachment 7: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
VI-S23-004	VI	\$23	Attachment 7: Comments Section	Instr.	Attachment 7: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
VI-S24-001	VI	\$24	MASTER PRODUCTION RECORD REVISION HISTORY	Instr.	The following is included for reference only and is not part of the executable record.	Version Change Reason	— (revision history) —	Excl
VI-S24-002	VI	\$24	MASTER PRODUCTION RECORD REVISION HISTORY	Instr.	This is a new master production record (MPR) for use in the downstream processing of AZD0543 Viral Inactivation.		— (revision history) —	Excl
VI-S24-003	VI	\$24	MASTER PRODUCTION RECORD REVISION HISTORY	Instr.	The information contained in this document is derived from technical report REP-0494621 and REP-0414101 and sampling and testing plan PLAN-0124866.		— (revision history) —	Excl
CDF-S1-001	CDF	\$1	Process Summary	Instr.	This master production record (MPR) describes the procedure for the Concentration, Diafiltration, Recovery and Dilution, of the AZD0543 product in the downstream process outlined below.		— reference / not recordable —	Refe
CDF-S2-001	CDF	\$2	Downstream Process Flow Sheet	Fields		MABR-0027568 / PN: 8012412 / AZD0543 - Centrifuge Harvest (GPF, 2000 L Scale, Process 1) MABR-0027569 / PN: 8012440 / AZD0543 - Affinity Chromatography (DurA Cycle A50, GPF, 2000 L Scale, Process 1)	— reference / not recordable —	Refe

						MABR-0027570 / PN: 8012441 / AZD0543 - Virus Inactivation (Low pH, GPF, 2000 L Scale, Process 1) MABR-0027573 / PN: 8012472 / AZD0543 - Anion Exchange Chromatography (CaptoQ, GPF, 2000 L Scale, Process 1) MABR-0027574 / PN: 8012473 / AZD0543 - Cation Exchange Chromatography (Poros HS50, GPF, 2000 L Scale, Process 1) MABR-0027575 / PN: 8012474 / AZD0543 - Virus Filtration (VPro, GPF, 2000 L Scale, Process 1) MABR-0027576 / PN: 8012475 / AZD0543 - Concentration and Diafiltration (Large Skid, GPF, 2000 L Scale, Process 1) MABR-0027577 / PN: 8012477 / AZD0543 - Formulation and Bulk Filtration (Cryovault, GPF, 2000 L Scale, Process 1) Storage Final Sterile Filtration and Vialing (off-site fill/finish manufacturer)		
CDF-S3-001	CDF	\$3	Referenced Documents	Instr.	FORM-0071789: Skid UDFD Cassette Installation and Pre-Sanitization Data Sheet		— reference / not recordable —	Refe
CDF-S3-002	CDF	\$3	Referenced Documents	Instr.	SOP-0080153: Operation of Formulation Portable Tank (TK-1927)		— reference / not recordable —	Refe
CDF-S3-003	CDF	\$3	Referenced Documents	Instr.	SOP-0080185: WFI Point of Use Operations		— reference / not recordable —	Refe
CDF-S3-004	CDF	\$3	Referenced Documents	Instr.	SOP-0080273: Operation of Advanced MicroOsmometer		— reference / not recordable —	Refe
CDF-S3-005	CDF	\$3	Referenced Documents	Instr.	SOP-0080295: DLIMS Sample Submission for Manufacturing Sciences		— reference / not recordable —	Refe
CDF-S3-006	CDF	\$3	Referenced Documents	Instr.	SOP-0080297: Benchtop pH and Conductivity Meter Operation		— reference / not recordable —	Refe
CDF-S3-007	CDF	\$3	Referenced Documents	Instr.	SOP-0080303: WFI Hold Tank Operation (TK-1458)		— reference / not recordable —	Refe
CDF-S3-008	CDF	\$3	Referenced Documents	Instr.	SOP-0080423: Operation and Maintenance of the LevTech and LevMixer Mixing Systems		— reference / not recordable —	Refe
CDF-S3-009	CDF	\$3	Referenced Documents	Instr.	SOP-0080509: Osmometer Operation		— reference / not recordable —	Refe
CDF-S3-010	CDF	\$3	Referenced Documents	Instr.	SOP-0080525: Operation and Exterior Cleaning of Agitator Controlled 100 L Portable Formulation Tanks: TK-2149 and TK-2150		— reference / not recordable —	Refe
CDF-S3-011	CDF	\$3	Referenced Documents	Instr.	SOP-0080539: Operation, Cleaning and Maintenance of Magnetic Stirplates		— reference / not recordable —	Refe
CDF-S3-012	CDF	\$3	Referenced Documents	Instr.	SOP-0080839: Peristaltic Pump Operation		— reference / not recordable —	Refe
CDF-S3-013	CDF	\$3	Referenced Documents	Instr.	SOP-0081194: Laminar Flow Cabinet Operation and Cleaning		— reference / not recordable —	Refe
CDF-S3-014	CDF	\$3	Referenced Documents	Instr.	SOP-0107056: cGMP Labeling Practices for Controlled Materials, Equipment and Product		— reference / not recordable —	Refe

CDF-S3-015	CDF	\$3	Referenced Documents	Instr.	SOP-0107091: Operation and Maintenance of the SoloVPE System for Protein Concentration Determination.		— reference / not recordable —	Refe
CDF-S3-016	CDF	\$3	Referenced Documents	Instr.	SOP-0107095: Operation of the Product Hold Fixed Tanks in the Gaithersburg Pilot Facility		— reference / not recordable —	Refe
CDF-S3-017	CDF	\$3	Referenced Documents	Instr.	SOP-0107097: Sampling from Process Vessels		— reference / not recordable —	Refe
CDF-S3-018	CDF	\$3	Referenced Documents	Instr.	SOP-0107102: Operation and Maintenance of the Integrity Magnetic Mixer Single-Use Mixing System		— reference / not recordable —	Refe
CDF-S3-019	CDF	\$3	Referenced Documents	Instr.	SOP-0107111: Ultrafiltration Skid Operation, Cleaning, and Storage		— reference / not recordable —	Refe
CDF-S3-020	CDF	\$3	Referenced Documents	Instr.	SOP-0107117: Thermometer Use		— reference / not recordable —	Refe
CDF-S3-021	CDF	\$3	Referenced Documents	Instr.	SOP-0107420: Pressure Hold Test		— reference / not recordable —	Refe
CDF-S3-022	CDF	\$3	Referenced Documents	Instr.	MABR-0027575 / PN: 8012474: AZD0543 – PN: 8012474 / AZD0543 - Virus Filtration (VPro, GPF, 2000 L Scale, Process 1)		— reference / not recordable —	Refe
CDF-S3-023	CDF	\$3	Referenced Documents	Instr.	MABR-0027577 / PN: 8012477: AZD0543 – PN: 8012477 / AZD0543 - Formulation and Bulk Filtration (Cryovault, GPF, 2000 L Scale, Process 1)		— reference / not recordable —	Refe
CDF-S4-001	CDF	\$4	Bill of Materials – Equipment and Components	Fields		Part No./ Equipment ID Quantity PALL Large Scale UF/DF Skid Mixer Drive Peristaltic Pump Stir Plate Conductivity Meter pH meter Thermometer Floor Scale/ Benchtop Balance Filter Integrity Tester Stainless Steel Dip tubes Sight glasses 5 L Glass Bottle 10 L Glass Bottle 20 L Glass Bottle Part No./ Equipment ID Quantity UFDF Membranes PELLICON 3 30kDa (C Screen) 0.11 m2 PELLICON 3 30kDa (C Screen) 0.57 m2 PELLICON 3 30kDa (C Screen) 1.14 m2 Pellicon Manifold Support Plate Portable Formulation Tanks 100 L Diafiltration / Concentration Tank 200 L Diafiltration / Concentration Tank 100 L Diafiltration / Concentration Tank 100 L Diafiltration / Concentration Tank Vent Filters for the Diafiltration / Concentration Tanks Filter – Aervent, 0.2 µm, Hydrophobic, PTFE Membrane (5 in.) Filter – Aervent, 0.2 µm, Hydrophobic PTFE Membrane (10 in.) Product Filters for Filtration Filter – 10-inch Opticap XLT Capsule w/ SHC Membrane, Sterile Filter- 10 in., Opticap XLT 10 Capsule Filter with Millipore Express SHC Hydrophilic Membrane, 0.5/0.2 µm Filter – SHC 10” 0.5+0.2 um Capsule with C-flex Filter – SHC 3” 0.5+0.22 um Capsule with C-flex	SMPL: Display BOM	Cove

						Filter – 30 in., Opticap□ XLT 30 Capsule Filter with Millipore Express□ SHC Hydrophilic Membrane, 0.5/0.2 □m Filter- 4 in., Opticap XL Capsule Filter with Hydrophilic Durapore Membrane, 0.22 □m Filter- Millipak 200 0.22 µm, Hydrophilic Gamma Irradiated Bags for Product Mixing/Collection LevTech Mix Bag (50 L) LevTech Mix Bag (100 L) LevTech Mix Bag (200 L) LevTech Mix Bag (300 L) LevTech Mix Bag (560 L) Bags for WFI Bags – Stedim 3D Bags with C-flex (300 L) Bags – Stedim 3D Bags with C-flex (200 L) Stedim Bag- 50 L Flexel 2D Bag with C-flex		
CDF-S4-002	CDF	\$4	Bill of Materials – Equipment and Components	Instr.	* = Materials with bolded filter part numbers require autoclaving prior to use		SMPL: Display BOM	Cove
CDF-S4-003	CDF	\$4	Bill of Materials – Equipment and Components	Instr.	Additional Manufacturing Supplies Used in Process:		SMPL: Display BOM	Cove
CDF-S4-004	CDF	\$4	Bill of Materials – Equipment and Components	Note	If additional materials (filter, bag, etc.) need to be used during processing:		SMPL: Display BOM	Cove
CDF-S4-005	CDF	\$4	Bill of Materials – Equipment and Components	Instr.	Prior to use, ensure the material is acceptable for use based on Section 4.		SMPL: Display BOM	Cove
CDF-S4-006	CDF	\$4	Bill of Materials – Equipment and Components	Instr.	Record the applicable material information.		SMPL: Display BOM	Cove
CDF-S4-007	CDF	\$4	Bill of Materials – Equipment and Components	Instr.	Document the reason for the change in Attachment 9: Comments Section.	Serial No., if applicable	SMPL: Display BOM	Cove
CDF-S5-001	CDF	\$5	Process Flowchart	Fields		WFI Flush prior to Equilibration Water for Injection (WFI) Equilibration 8012555: 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 Concentration 1 of the Virus Filtration (VF) Product (8012474) to 100 g/L Diafiltration Concentration 2 to 210 g/L Recovery flush to 190 g/L Dilution to 180 g/L Filtration, Sampling and Storage System Sanitization A0013-D: 0.5 N Sodium Hydroxide	SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S5-002	CDF	\$5	Process Flowchart	Instr.	Sanitization buffer batch number and exp date are documented in Step 20.3.	System Storage A0014-D: 0.1 Sodium Hydroxide Storage buffer batch number and exp. date are documented in Step 20.3. Recorded by/ Date	SMPL: Additional Solution Batches (Solution Switch)	Cove

CDF-S5-003	CDF	\$5	Process Flowchart	Instr.	Additional Solution Batches Used in Process		SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S5-004	CDF	\$5	Process Flowchart	Note	If a solution (of a different batch noted in Section 5) needs to be replaced during processing:		SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S5-005	CDF	\$5	Process Flowchart	Instr.	Indicate the switch at the step it is occurring. Ensure the new solution is the same part number as the solution being replaced.		SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S5-006	CDF	\$5	Process Flowchart	Instr.	Record the solution part number, batch number (and/or process order number) and expiration date below.		SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S5-007	CDF	\$5	Process Flowchart	Instr.	Document the change in Attachment 9: Comments Section.	Applicable Step Number Part Number Batch Number	SMPL: Additional Solution Batches (Solution Switch)	Cove
CDF-S6-001	CDF	\$6	Process Notes	Note	All activities are performed at 15 °C to 27 °C.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-002	CDF	\$6	Process Notes	Note	For all calculations, assume 1 L = 1 kg and 1 mL = 1 g unless otherwise noted.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-003	CDF	\$6	Process Notes	Note	Verify that all solutions, raw materials, and standards are within expiration prior to use.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-004	CDF	\$6	Process Notes	Note	When an equipment ID is written in the MPR, the operator is indicating they have visually inspected and verified calibration prior to use.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-005	CDF	\$6	Process Notes	Note	If additional mixing/recirculation is required after initial sampling of the product, record the mixing information in Attachment 9: Comments Section.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-006	CDF	\$6	Process Notes	Note	CAUTION: Sodium Hydroxide is corrosive. A face shield must be worn when the solution is being connected and/or disconnected from the UF/DF system.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-007	CDF	\$6	Process Notes	Note	If additional WFI is pulled from the WFI drop during processing, record the WFI valve ID in Attachment 9: Comments Section.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S6-008	CDF	\$6	Process Notes	Note	If a non-routine sample is requested by memo, complete Attachment 4: Non-Routine Sampling Record.		SMPL: Process Notes & Global Limits (Acknowledgement)	Cove
CDF-S7-001	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Record the room number where this process will occur.	Room Number	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-002	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Obtain and record the following information from the AZD0543 CIPDS (FORM-0071789) and Skid Logbook.	CIPDS part number CIPDS lot number Skid ID Cassette installation date No. of cassettes used Cassette part number Cassette batch number Cassette expiration date Additional cassette batch number, if applicable Additional cassette expiration date, if applicable Hold-up volume of the UF/DF Skid L=kg Total surface area of cassettes Support Plate part number Support Plate batch number Plate #1	SMPL: Minimum Membrane Surface Area (block stack)	Cove

						Plate #2 Support Plate expiration date Plate #1 Plate #2 Logbook Number CIPDS SAP Consumption Performed by/Date		
CDF-S7-003	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Pre-Processing Activities for the UF/DF Skid Verify the data for the UF/DF Skid to be used. From the logbook, record in Step 7.4 the logbook number, the CIPDS batch number, and the date when the actions were documented. Ensure that the UF/DF Skid's most immediate storage after sanitization has not exceeded 30 days storage. If the storage duration is greater than 30 days, contact a supervisor and document actions taken in Attachment 9: Comments Section.	Pre-Processing Activities for the UF/DF Skid Action Logbook Number CIPDS Batch Number Date Sanitization NWP Storage DF-1244 Skid Storage Duration (≤ 30 days)	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-004	CDF	\$7	Initial Product and UF Membrane Information	Instr.	In the Step 7.6 "Virus Filtration Product Information" table: If a tank was used, N/A Vessels 2 and 3. For bag(s), N/A any unused Vessel fields. Record the net weight of each Virus Filtration Product from (MABR-0027575) MPR or the label of each product transport vessel. Record the tare weight of each Virus Filtration Product Vessel. Record the product concentration of each Virus Filtration Product Vessel. Record the corresponding DLIMS project number and sample number for the Virus Filtration Product(s) vessels.	Virus Filtration Product Information Vessel 1 Vessel 2, if applicable Vessel 3, if applicable Virus Filtration Product Net Weight Tare Weight VF Product Concentration DLIMS Sample Number DLIMS Project Number Recorded By/Date	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-005	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Calculate the Virus Filtration Product volume in each vessel. NOTE: N/A any unused Vessel fields below.	Virus Filtration Product Net Weight (Step 7.6) Density of VF Product Virus Filtration (VF) Product Volume Vessel 1 1.014 kg/L Vessel 1 1.014 kg/L Vessel 2, if applicable 1.014 kg/L Vessel 2, if applicable 1.014 kg/L Vessel 3, if applicable 1.014 kg/L Vessel 3, if applicable 1.014 kg/L	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-006	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Calculate the total grams of Virus Filtration Product in each vessel. NOTE: N/A any unused Vessel fields below.	Virus Filtration (VF) Product Volume (Step 7.7) Virus Filtration Product Concentration (Step 7.6) Total grams of VF Product per Vessel Vessel 1 Vessel 1 Vessel 1 Vessel 2, if applicable Vessel 2, if applicable Vessel 2, if applicable Vessel 3, if applicable Vessel 3, if applicable Vessel 3, if applicable Total grams of VF Product before sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-007	CDF	\$7	Initial Product and UF Membrane Information	Instr.	Calculate the total membrane surface area needed for the UFDF process:	Total grams of VF Product before sampling (Step 7.8) Maximum Load (g of Product / m2) Minimum membrane surface area	SMPL: Minimum Membrane Surface Area (block stack)	Cove
CDF-S7-008	CDF	\$7	Initial Product and UF Membrane Information	Note	Verify the total installed surface area of cassettes (Step 7.2) is greater than or equal to the minimum membrane surface area calculated in Step 7.9.		SMPL: Minimum Membrane Surface Area (block stack)	Cove

CDF-S8-001	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Set up per SOP-0107111. WFI collected/obtained directly from the WFI Hold Tank (TK-1458) per SOP-0080303: Verify the SIP of the WFI Hold Tank was completed and the tank status is "Clean." Before connecting to the UF skid, WFI Flush the TK-1458 WFI port on the transfer panel per the following instructions: Ensure an appropriately sized Tee with bleed valve is connected to the manual valve leading from TK-1458 on the transfer panel, then attach a clean hose to the other end of the Tee. Connect the other end of the hose to a process drain port. Ensure any valves on the process drain port are open, if applicable. Open the manual valve leading from the TK-1458 port. Ensure the valve is completely open. Flush TK-1458 through the transfer panel port manual valve to process drain for ≥30 seconds, then close the manual valve on the TK-1458 transfer panel port. N/A the bag information. WFI was/will be collected into a bag: Record the WFI collection vessel information. Verify that the WFI port has been flushed per SOP-0080185. Charge into the vessel and record the charge information.	Bag, if applicable WFI Valve ID/Tank ID WFI Charge Date/Time Date: Time: WFI Expiration Date/Time* Date: Time:	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-002	CDF	\$8	WFI Flush prior to Equilibration	Instr.	* The WFI Expiration Date/Time is 24 hours from the WFI Charge Date/Time.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-003	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Confirm that the pH meter has been standardized per SOP-0080297. Confirm that the conductivity meter has been standardized per SOP-0080297.	Meter ID	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-004	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Confirm that the thermometer is within its calibration interval. Record the equipment ID of the thermometer used.	Thermometer ID	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-005	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Ensure that the Ultrafiltration (UF) system is set up for WFI flush of the installed cassettes according to SOP-0107111.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-006	CDF	\$8	WFI Flush prior to Equilibration	Instr.	On the UF/DF Skid HMI, enter the total installed cassette surface area in m2 (Step 7.2): Click on the membrane Area Setpoint input box. Enter Total Surface Area of Cassettes in m2 (Step 7.2) and press ENTER. Click ACCEPT on the Membrane Area Change pop-up window.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-007	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Connect the WFI supply inlet to the UF/DF Skid feed line. Ensure that the retentate and permeate lines on the UF/DF Skid are directed to waste.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-008	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Calculate the target volume of solution needed for a 22.0 L/m2 WFI and Equilibration flushes.	Target Flush Volume Total Surface Area of Cassettes (Step 7.2) Target WFI and Equilibration Flush Volume	SMPL: Run Skid Recipe (block stack)	Cove

CDF-S8-009	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_PrepWFI_Batch number_Processing Date NOTE: The membranes should be flushed no more than 24 hours before use.	Batch ID Start Time	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-010	CDF	\$8	WFI Flush prior to Equilibration	Instr.	At the user prompt, input the flush volume set points per the table below.	Process Step	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-011	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Input Title	Entered Set Point Permeate Flush PERMEATE_FLUSH_VOL ≥ Step 8.7 Retentate Flush RET_FLUSH_VOL ≥ Step 8.7	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-012	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the pre-use WFI flush. Perform the WFI flush of the UF/DF Skid per SOP-0107111. Sample and document the flush conductivities per Step 8.12.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-013	CDF	\$8	WFI Flush prior to Equilibration	Instr.	At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-014	CDF	\$8	WFI Flush prior to Equilibration	Instr.	At the end of the recipe, take a sample of the retentate and permeate and record the results below.	Retentate Values Conductivity (≤ 1 mS/cm) mS/cm Temperature (15 – 25°C) Permeate Values Conductivity (≤ 1 mS/cm) mS/cm Temperature (15 – 25°C)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-015	CDF	\$8	WFI Flush prior to Equilibration	Note	If the conductivity is in range, N/A the remainder of Section 8 and go to Section 9. If the conductivity is not in range, go to the next step.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-016	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_PrepWFI2_Batch Number_Processing Date. NOTE: The membranes should be flushed no more than 24 hours prior to use.	Batch ID Start Time At the user prompt, input the flush volumes per the table below. Process Step	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-017	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Input Title	Entered Set Point Permeate Flush PERMEATE_FLUSH_VOL ≥ Step 8.7 Retentate Flush RET_FLUSH_VOL ≥ Step 8.7	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-018	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the WFI flush of the UF/DF Skid. Perform the WFI flush of the UF/DF Skid per SOP-0107111. Sample and document the flush conductivities per Step 8.18.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-019	CDF	\$8	WFI Flush prior to Equilibration	Instr.	At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S8-020	CDF	\$8	WFI Flush prior to Equilibration	Instr.	Take a sample of the retentate and permeate and record the results below. If the conductivities are not within the	Retentate Values Conductivity (≤ 1 mS/cm) mS/cm Temperature (15 – 25°C) Permeate Values Conductivity	SMPL: Run Skid Recipe (block stack)	Cove

					specified range after repeating the recipe, contact an area supervisor and document actions taken in Attachment 9: Comments Section.	(≤ 1 mS/cm) mS/cm Temperature (15 – 25°C)		
CDF-S9-001	CDF	\$9	End of Virus Filtration Product Storage	Note	If the Virus Filtration Product was warmed up (from cold storage), ensure the time the product was removed from cold storage is recorded in Attachment 5: Hold Times.		SMPL: Product Mixing	Cove
CDF-S9-002	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Mix the Virus Filtration product for at least 15 minutes and record mixing information below: Follow SOP-0080423 for the LevTech mixer. Follow SOP-0107095 for agitation in fixed tanks. If mixing by recirculation, N/A this step and document mixing in Attachment 7: Product Recirculation Work Sheet. N/A any unused Vessel fields below.	Vessel 1 Mixer ID (if applicable) Agitation Speed Start Time End Time Duration (≥ 15 min) Vessel 2, if applicable Mixer ID (if applicable) Agitation Speed Start Time End Time Duration (≥ 15 min) Vessel 3, if applicable Mixer ID (if applicable) Agitation Speed Start Time End Time Duration (≥ 15 min)	SMPL: Product Mixing	Cove
CDF-S9-003	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Remove at least the sample volume per Attachment 3: Sampling Record from each of the product collection vessel per SOP-0107097. If non-routine samples are requested by memo, complete Attachment 4: Non-Routine Sampling Record. N/A any unused Vessel fields below.	Sampling Date and Time, Vessel 1 Date: Time: Sampling Date and Time, Vessel 2, if applicable Date: Time: Sampling Date and Time, Vessel 3, if applicable Date: Time:	SMPL: Product Mixing	Cove
CDF-S9-004	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Aliquot the sample (sample designation: 8012475-10-1, 8012475-10-2, or 8012475-10-3, as applicable) per Attachment 3: Sampling Record. If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample per SOP-0107056.		SMPL: Product Mixing	Cove
CDF-S9-005	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Submit the sample to the storage location per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.		SMPL: Product Mixing	Cove
CDF-S9-006	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Place the remaining sample in a 50 ml conical tube and measure the temperature of the sample per SOP-0107117. Record the reading below. N/A any unused Vessel fields below.	Vessel 1 Temperature Vessel 2 Temperature, if applicable Vessel 3 Temperature, if applicable	SMPL: Product Mixing	Cove
CDF-S9-007	CDF	\$9	End of Virus Filtration Product Storage	Note	If the VF product is in a bag or portable tank, N/A this step. Otherwise, continue below. In Purification, ensure the following: All the necessary transfer panel connections are made for transfer of the VF product to Formulation per SOP-0107099. The condensate flex hose on the transfer panel is removed and the manual valve that the flex hose was attached to is closed. Use SOP-0080548 for Large Purification fixed tanks and SOP-0080323 for Small Purification fixed tanks.		SMPL: Product Mixing	Cove
CDF-S9-008	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Weigh the product vessel(s) and record below. N/A any unused Vessel fields. Calculate the Net Weight for each vessel. The "Total Net Weight after sampling" is the sum of the net weights for each vessel.	Vessel 2, if applicable 3, if applicable Gross Weight Virus Filtration Product Vessel Tare Weight (Step 7.6) Net Weight a Total Net Weight of VF Product after sampling (Vessel 1 + Vessel 2 + Vessel 3) Scale/Balance ID Cal. Due Date Net Weight = Gross	SMPL: Product Mixing	Cove

						weight – Virus Filtration Product Vessel Tare Weight		
CDF-S9-009	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Calculate the Virus Filtration Product Volume after sampling. NOTE: N/A any unused Vessel fields below.	Net Weight (Step 9.8) Density of VF Product Virus Filtration Product Volume after sampling Vessel 1 1.014 kg/L Vessel 1 1.014 kg/L Vessel 2, if applicable 1.014 kg/L Vessel 2, if applicable 1.014 kg/L Vessel 3, if applicable 1.014 kg/L Vessel 3, if applicable 1.014 kg/L	SMPL: Product Mixing	Cove
CDF-S9-010	CDF	\$9	End of Virus Filtration Product Storage	Instr.	Calculate the Total Grams of Virus Filtration Product after sampling. NOTE: N/A any unused Vessel fields below.	Virus Filtration Product Volume after sampling (Step 9.9) Virus Filtration Product concentration (Step 7.6) Total Grams of Virus Filtration Product after sampling per Vessel Vessel 1 Vessel 1 Vessel 1 Vessel 2, if applicable Vessel 2, if applicable Vessel 2, if applicable Vessel 3, if applicable Vessel 3, if applicable Vessel 3, if applicable Total grams of Virus Filtration Product after sampling (Vessel 1 grams + Vessel 2 grams + Vessel 3 grams)	SMPL: Product Mixing	Cove
CDF-S10-001	CDF	\$10	Equilibration	Instr.	After the skid WFI flush is completed and prior to proceeding to Equilibration, standardized the skid's UV meter per SOP-0107111.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-002	CDF	\$10	Equilibration	Instr.	Obtain the equilibration buffer, 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 (PN: 8012555). Connect the equilibration buffer to the UF/DF Skid feed line.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-003	CDF	\$10	Equilibration	Instr.	Remove a sample of the equilibration buffer and record the results below.	Conductivity mS/cm Temperature (15 - 25°C)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-004	CDF	\$10	Equilibration	Instr.	Download and run DF1244_EQUIL according to SOP-0107111 to flush the UF membrane(s). Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_Equil_Batch Number_Processing Date.	Batch ID Start Time At the user prompt, input the values per the table below. Process Step	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-005	CDF	\$10	Equilibration	Instr.	Input Title	Entered Set Point Permeate Flush PERMEATE_FLUSH_VOL ≥ Step 8.7 Retentate Flush RET_FLUSH_VOL ≥ Step 8.7 Trans-Membrane Pressure (psid) EQUIL_TMP Cross Flow (L/min/ft2) c EQUIL_Crossflow The TMP can be set as needed up to NOR of ≤ 35 psid. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure does not exceed NOR of ≤ 60 psig. Cross flow can be adjusted within the NOR range of 0.3 – 0.7 L/min/ft2 as needed to ensure feed pressure does not exceed NOR of ≤ 60 psig. If the cross flow is adjusted, record the new value in	SMPL: Run Skid Recipe (block stack)	Cove

						Attachment 9: Comments Section. The UF/DF Skid measures the cross flow in L/min/ft2. For conversion reference, L/min/ft2 = L/min/m2 10		
CDF-S10-006	CDF	\$10	Equilibration	Instr.	Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the equilibration of the UF/DF Skid. NOTE: Ensure the pH meets the range required per Step 10.8.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-007	CDF	\$10	Equilibration	Instr.	At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops. Record the amount of equilibration buffer 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 flushed using the UF/DF skid's retentate and permeate totalizers. NOTE: The totalizer HMI tags are FQI-301 (retentate) and FQI-201 (permeate).	Retentate Totalizer (FQI-301) Permeate Totalizer (FQI-201)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-008	CDF	\$10	Equilibration	Instr.	At the end of the recipe, take a sample of the permeate and record the results below.	Permeate Values Conductivity mS/cm Temperature (15 - 25°C)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-009	CDF	\$10	Equilibration	Note	If the pH is in range, mark Steps 10.10 through 10.15 as N/A, then proceed to Section 11. If the pH is not in range, go to the next step.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-010	CDF	\$10	Equilibration	Instr.	Download and run DF1244_Equil according to SOP-0107111 to flush the UF membrane(s). Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_Equil2_Batch Number_Processing Date.	Batch ID Start Time At the user prompt, input the values per the table below. Process Step	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-011	CDF	\$10	Equilibration	Instr.	Input Title	Entered Set Point Permeate Flush PERMEATE_FLUSH_VOL ≥ Step 8.7 Retentate Flush RET_FLUSH_VOL ≥ Step 8.7 Trans-Membrane Pressure (psid) EQUIL_TMP Cross Flow (L/min/ft2) c EQUIL_Crossflow The TMP can be set as needed up to NOR of ≤ 35 psid. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure does not exceed NOR of ≤ 60 psig. Cross flow can be adjusted within the NOR range of 0.3 – 0.7 L/min/ft2 as needed to ensure feed pressure does not exceed NOR of ≤ 60 psig. If the cross flow is adjusted, record the new value in Attachment 9: Comments Section. The UF/DF Skid measures the cross flow in L/min/ft2. For conversion reference, L/min/ft2 = L/min/m2 10	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-012	CDF	\$10	Equilibration	Instr.	Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the equilibration of the UF/DF Skid. NOTE: Ensure the pH meets the range required per Step 10.14.		SMPL: Run Skid Recipe (block stack)	Cove

CDF-S10-013	CDF	\$10	Equilibration	Instr.	At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops. Record the amount of equilibration buffer 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9 flushed using the UF/DF skid's retentate and permeate totalizers. NOTE: The totalizer HMI tags are FQI-301 (retentate) and FQI-201 (permeate).	Retentate Totalizer (FQI-301) Permeate Totalizer (FQI-201)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-014	CDF	\$10	Equilibration	Instr.	At the end of the recipe, take a sample from the permeate. Record the results below.	Permeate Values Conductivity mS/cm Temperature (15 - 25°C)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S10-015	CDF	\$10	Equilibration	Note	If the pH is in range, go to the next step. If the pH is not in range, contact an area supervisor and document actions in Attachment 9: Comments Section.	Supervisor Contacted	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S11-001	CDF	\$11	Vessel Set-up	Instr.	Obtain the Retentate Vessel. Label the vessel "Retentate Vessel". Record the vessel's tank ID, clean expiration date and time, sterilization expiration date and time, Cleaning and SIP Batch IDs and date of passed pressure test. Record the volume of the Retentate Vessel.	Tank ID Cleaning Exp. Date / Time Tank Date of Passed Pressure Test Cleaning Batch ID Sterilization Exp. Date/ Time Sterilization Batch ID Retentate Vessel Volume Vent Filter Information Part Number Batch Number Expiration Date Serial Number	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-002	CDF	\$11	Vessel Set-up	Note	Verify that the vent filter is open on the Retentate Vessel.		SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-003	CDF	\$11	Vessel Set-up	Instr.	Retentate Vessel Tare Set-Up Label the manual valve on the dip tube of the tank as valve A. Endcap and clamp valve A. Label the tank outlet valve as valve B. End cap the elbow connected to valve B. Label the valve on the tank's retentate tube as valve C. End cap valve C.		SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-004	CDF	\$11	Vessel Set-up	Instr.	Ensure that the Retentate Tank vent filter is open per Step 11.2, then open Valves A, B, and C all the way (while they are end capped) to relieve pressure between the endcap and the valve outlets. Then, close Valves A, B, and C.		SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-005	CDF	\$11	Vessel Set-up	Instr.	Record the tare weight of the vessel before it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.	Scale / Balance ID Scale/Balance Calibration Due Date Tare weight of Retentate Vessel before attachment	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-006	CDF	\$11	Vessel Set-up	Instr.	Per Attachment 6: Retentate Tank , ensure a sight glass is installed on the following: On the VF Product transfer panel manual valve outlet (valve H) If the VF Product is in a bag, N/A the "Transfer Panel (Valve H)" sight glass fields in this step. Prior to valve A (which is attached to the Retentate Vessel inlet). Prior to valve C (which is attached to the retentate tube on the Retentate Vessel). Record the sight glass information below.	Sight Glass Information	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-007	CDF	\$11	Vessel Set-up	Instr.	Transfer Panel (Valve H), if applicable	Valve A Valve C	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove

CDF-S11-008	CDF	\$11	Vessel Set-up	Instr.	Retentate Vessel and UF/DF System Set-Up Set up the Retentate Vessel and DF-1244 system per Attachment 6: Retentate Tank and DF-1244 Set-up. Record the tubing/hose information in Attachment 1: Process Tubing/Hose Information. For tubing, record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. For hoses, record the hose serial number. NOTE: If hoses are used and a serial number is not available, just record that a hose was used in Attachment 1: Process Tubing/Hose Information. Obtain elbows, tees, and manual valves as needed and connect these components to the Retentate Vessel and the UF/DF skid per Attachment 6: Retentate Tank and DF-1244 Set-up. Ensure that manual valves A, B, C, D, E, F, G, H, and I are labeled. Label the Permeate, Retentate, Feed, Recovery, and Charge/Diafiltration (DF) lines.		SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-009	CDF	\$11	Vessel Set-up	Instr.	Prepare for system priming. Ensure the following valve states:	Valve States A, B, C, D, E, F, G, H, I	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-010	CDF	\$11	Vessel Set-up	Instr.	Ensure that valves E and H are closed. Initiate the transfer phase on the PCS HMI for the VF Product and all the buffers and the that are coming through the transfer panel. NOTE: Confirm that connections are made in Large Purification before proceeding.	System Prime: Recovery Line Prime the Recovery Line: Ensure valves A, B, C, D and F are closed. Open bleed valve B-2. Ensure the tubing is directed to waste. Open valves E and G to prime the Recovery Line with the DF Buffer. If necessary, use a pump to prime the Recovery Line. Once no more air is observed exiting bleed valve B-2, close valve E then close bleed valve B-2. Close valve G. NOTE: Buffer charging is performed using diafiltration buffer (PN: 8012555 : 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9) System Prime: Charge/Diafiltration (DF) Line Prime the Charge/DF Line: Ensure valves A, B, C, D, and I are closed. Open bleed valve B-1. Ensure the tubing is directed to waste. Open valves E and F from the Diafiltration Buffer to the Charge/DF Line. Use the pump to prime the Charge/DF Line, as necessary. Once no more air is observed exiting bleed valve B-1, close valve E, then close bleed valve B-1. Close valve F. NOTE: Buffer charging is performed using diafiltration buffer (PN: 8012555 : 20 mM histidine/histidine-HCl, 120 mM arginine-HCl, pH 5.9)	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S11-011	CDF	\$11	Vessel Set-up	Instr.	Record the tare weight of the vessel after it is attached to the UF/DF System. Record the scale/balance equipment ID and calibration due date.	Scale / Balance ID Scale / Balance Calibration Due Date Tare weight of Retentate Vessel after attachment Secure the wheels of the Retentate	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove

						Vessel. Tare (zero) the scale holding the Retentate Vessel.		
CDF-S12-001	CDF	\$12	Process Input Calculations	Instr.	Calculate the volume at 80% Retentate Vessel Capacity.	Retentate Vessel Volume (Step 11.1) Capacity Limit Retentate Vessel Charge Volume/Weight at 80% Capacity L = kg	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-002	CDF	\$12	Process Input Calculations	Instr.	Calculate the volume at 40% Retentate Vessel Capacity.	Retentate Vessel Volume (Step 11.1) Buffer Charge Target Retentate Vessel Charge Volume/Weight at 40% Capacity L = kg	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-003	CDF	\$12	Process Input Calculations	Instr.	Calculate the product volume in the UF/DF Skid after Concentration 1.	Total Grams of VF Product after sampling (Step 9.10) Concentration 1 Range Product volume in the UF/DF System after Concentration 1 110 g/L (max) L (Min) 100 g/L (Target) L (Target) 90 g/L (min) L (Max)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-004	CDF	\$12	Process Input Calculations	Instr.	Calculate the product volume in the Retentate Vessel after Concentration 1.	Product volume in the UF/DF System after Concentration 1 (Step 12.3) Hold-up volume of the UF/DF Skid (Step 7.2) Product volume in the Retentate Vessel after Concentration 1 L (Min) L (Min) L (Target) L (Target) L (Max) L (Max)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-005	CDF	\$12	Process Input Calculations	Instr.	Calculate the product weight in the Retentate Vessel after Concentration 1.	Product volume in the Retentate Vessel after Concentration 1 (Step 12.4) Density of the Concentration 1 Product Product weight in the Retentate Vessel after Concentration 1 L (Min) kg/L kg (Min) L (Target) kg/L kg (Target) L (Max) kg/L kg (Max)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-006	CDF	\$12	Process Input Calculations	Instr.	Calculate the permeate volume range for Diafiltration.	Diafiltration Volume Range (DVs) Product Volume in the UF/DF System after Concentration 1 (Step 12.3) Permeate volume range after Diafiltration L (Min) L (Target)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-007	CDF	\$12	Process Input Calculations	Instr.	Calculate the product volume range in the UF/DF System after Concentration 2	Total Grams of VF Product after sampling (Step 9.10) Concentration 2 Range Product volume in the UF/DF System after Concentration 2 210 g/L (target) L (Target) g/L (min) L (Max)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-008	CDF	\$12	Process Input Calculations	Instr.	Calculate the product volume range in the Retentate Vessel after Concentration 2.	Product volume in the UF/DF System after Concentration 2 (Step 12.7) Hold-up volume of the UF/DF Skid (Step 7.2) Product volume in the Retentate Vessel after Concentration 2 L (Min) L (Min) L (Target) L (Target) L (Max) L (Max)	SMPL: Process Input Calculations (calc-block stack)	Cove
CDF-S12-009	CDF	\$12	Process Input Calculations	Instr.	Calculate the net weight range of the product in the Retentate Vessel after Concentration 2.	Product volume in the Retentate Vessel after Concentration 2 (Step 12.8) Density of the product after Concentration 2 at target Net Weight of product in the Retentate Vessel after Concentration 2 L (Min) kg/L kg	SMPL: Process Input Calculations (calc-block stack)	Cove

						(Min) L (Target) kg/L kg (Target) L (Max) kg/L kg (Max)		
CDF-S13-001	CDF	\$13	Concentration 1	Instr.	Ensure the Retentate Vessel is in the same configuration as it was when its Tare Weight after attachment was obtained in Step 11.12.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-002	CDF	\$13	Concentration 1	Instr.	To prepare for buffer charge, verify the following valve states on the Retentate Vessel.	A, B, C, D, E, F, G, H, I	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-003	CDF	\$13	Concentration 1	Instr.	Ensure that skid's hydraulic valves to the feed (XV101), retentate (XV301 and XV302) and permeate (XV201) lines are closed.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-004	CDF	\$13	Concentration 1	Instr.	Open valves A and F. Slowly open valve E from (PN: 8012555) and charge the Retentate Vessel with buffer to achieve the Vessel Charging Volume at 40% (Step 12.2). Once the weight from Step 12.2 has been reached, immediately close valves A, E, and F. • Document the Weight of the Retentate Vessel after buffer charge below.	Scale ID Cal. Due Date Weight of Retentate Vessel after buffer charge	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-005	CDF	\$13	Concentration 1	Instr.	To prepare for recirculation, ensure the following:	A, D, E, F, G, H, I	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-006	CDF	\$13	Concentration 1	Instr.	On the UF/DF Skid HMI: Open the valves to the feed (XV101), and retentate (XV301 and XV302). Ensure the permeate valve (XV201) is closed. Ensure all drain valves (XV103, XV102, and XV202) are closed. NOTE: Drain valves XV102, XV103, and XV202 should not be opened at any time during processing.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-007	CDF	\$13	Concentration 1	Instr.	Recirculate the charging buffer, click on the "Semi-Automatic" mode on the UF/DF skid screen and input the following initial parameters in the order listed:	Feed Pressure (psig) Feed Valve	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-008	CDF	\$13	Concentration 1	Instr.	Open	Retentate Valves	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-009	CDF	\$13	Concentration 1	Instr.	Open	Retentate Valves	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-010	CDF	\$13	Concentration 1	Instr.	Open	Permeate Valve	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-011	CDF	\$13	Concentration 1	Instr.	Drain Valves		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-012	CDF	\$13	Concentration 1	Instr.	Drain Valves		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-013	CDF	\$13	Concentration 1	Instr.	Drain Valves		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-014	CDF	\$13	Concentration 1	Instr.	The feed pressure can be adjusted as needed to maintain a target TMP of 20 psid and a maximum AL feed pressure ≤ 60 psig. Record changes to transmembrane pressure and/or cross flow setpoint(s) in Attachment 2: Operation		SMPL: Run Skid Recipe (block stack)	Cove

					Parameters. Record any additional changes in Attachment 9: Comments Section.			
CDF-S13-015	CDF	\$13	Concentration 1	Instr.	Record the start time as the time when the is pump turned on.	Start Time	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-016	CDF	\$13	Concentration 1	Instr.	Recirculate for at least 5 minutes. At the completion of recirculation, stop the skid pump and record the end time and duration below.	End time for recirculation Duration of recirculation (≥ 5 minutes) *	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-017	CDF	\$13	Concentration 1	Instr.	* NOTE: The duration is the time that elapses between the start time for recirculation (Step 13.8) and the end time for recirculation (this step).		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-018	CDF	\$13	Concentration 1	Instr.	Once buffer recirculation is complete, ensure the following valves on the Retentate Vessel are closed per the table below.	A, B, C, D, E, F, G, H, I	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-019	CDF	\$13	Concentration 1	Instr.	Ensure that valves E and F from the charge buffer (PN: 8012555) vessel are closed. Ensure that the valve A, H, and I on the Charge/DF Line are closed. Ensure valves to feed (XV101) and retentate (XV301 and XV302) are open. Ensure the permeate (XV201) line is closed.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-020	CDF	\$13	Concentration 1	Instr.	Charge the Retentate Vessel with the appropriate amount of Virus Filtered Product according to the charging condition below. Step 9.8: Total VF Product Net Weight after Sampling Step 12.2: Retentate Vessel Charge Volume/Weight at 40% Capacity	Condition Check one Action to take Step 12.2 ≥ Step 9.8	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-021	CDF	\$13	Concentration 1	Instr.	Charge in all the VF product (Step 9.8)	Step 12.2 ≤ Step 9.8	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-022	CDF	\$13	Concentration 1	Instr.	Charge to the Retentate Vessel Charge Volume/Weight at 80% Capacity (Step 12.1)		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-023	CDF	\$13	Concentration 1	Instr.	Prior to charging in the VF product, ensure the VF Product line and the Charge/DF Line are bled of air by ensuring the following valve states:	A, B, C, D, E, F, G H, I, B1	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-024	CDF	\$13	Concentration 1	Instr.	Once no more air is observed exiting from B-1, close B-1 and then close valves H and I. Open valve A.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-025	CDF	\$13	Concentration 1	Instr.	Open the valve H, then slowly open valve I to charge the appropriate weight of VF product into the Retentate Vessel per Step 13.12, then close valves H and I. Record the weight of the Retentate Vessel after charging and the scale information below. Record the start time of product charge below.	Start Time of Product Charge Weight of the Retentate Vessel after product charge Scale/Balance ID Scale/Balance Calibration Due Date	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-026	CDF	\$13	Concentration 1	Instr.	Turn on the agitator. Record the agitator speed setting. Adjust the agitation speed as necessary minimizing foaming. Record changes in speed in Attachment 2: Operation Parameters, and rationale for change in Attachment 9: Comments Section.	Agitator Setting	SMPL: Run Skid Recipe (block stack)	Cove

CDF-S13-027	CDF	\$13	Concentration 1	Instr.	Prior to product recirculation, ensure the following valve states:	A, D, E, F, G, H, I	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-028	CDF	\$13	Concentration 1	Instr.	On the skid HMI: Open the valves to the feed (XV101), and Retentate (XV301 and XV302). Ensure the permeate valve (XV201) is closed. Ensure all drain valves (XV103, XV102, and XV202) are closed. NOTE: Drain Valves XV102, XV103, and XV202 should not be opened at any time during processing.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-029	CDF	\$13	Concentration 1	Instr.	To recirculate the charged product, click on the "Semi-Automatic" mode on the UF/DF Skid screen and input the following initial parameters in the order listed: The feed pressure can be adjusted as needed to maintain a target TMP of 20 psid and a maximum feed pressure of ≤ 60 psig. Record changes to TMP and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section. Max feed pressure NOR is ≤ 60 psig.	The start time for recirculation is the time the pump turned on. Start Time	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-030	CDF	\$13	Concentration 1	Instr.	Continue to recirculate for at least five (5) minutes. Record the end time and the duration of recirculation. Stop the pump. Ensure valve D is closed.	End time for recirculation Duration of recirculation (≥ 5 minutes) *	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-031	CDF	\$13	Concentration 1	Instr.	* NOTE: The duration is the time that elapses between the start time for recirculation (Step 13.19) and the end time for recirculation (this step).		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-032	CDF	\$13	Concentration 1	Instr.	To prepare for Concentration 1: Open the permeate valve (XV201) on the skid HMI. Ensure the manual valves on the Retentate Vessel are open or closed per the table below.	D, E, F, G, I A, B, C, H	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-033	CDF	\$13	Concentration 1	Instr.	Click on the "Add Recipe" button in Recipe Control screen. Find and Select DF1244_CONC from the list. Record the Concentration 1 batch ID and start time from the system overview screen. The batch ID should follow this format: MPR Part Number_Conc_1 _Product Lot Number_Processing Date.	Batch ID Start Time At the user prompt, insert the following values. Process Step	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-034	CDF	\$13	Concentration 1	Instr.	Input Title	Set Points Trans-Membrane Pressure (psid) Initial_Conc_TMP Cross Flow (L/min/ft2) Initial_Conc_Crossflow	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-035	CDF	\$13	Concentration 1	Instr.	The transmembrane pressure can be set up to ≤ 35 psid as needed. If the value is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new value in Attachment 9: Comments Section.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-036	CDF	\$13	Concentration 1	Instr.	Push START to initiate the Concentration 1 process. The Start Time is the time the pump turns on.	Start Time	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-037	CDF	\$13	Concentration 1	Instr.	Concentration 1 Once the skid pump starts, carefully adjust manual valve I on the VF Product Line or adjust the	Maximum TMP (psid) Feed Flow (L/min/ft2) Feed Pressure (psig) UV	SMPL: Run Skid Recipe (block stack)	Cove

					pump speed to control the flow of VF Product into the Retentate Vessel. Maintain the gross charging weight from Step 13.14 until the Virus Filtration Product vessel(s) is empty. Adjust the crossflow and transmembrane pressure (TMP) setpoint(s) from their initial values, as necessary. See Step 13.23, for appropriate ranges. NOTE: If adjustment occurs, record the adjusted values, the time of adjustment, and the step number in Attachment 2: Operation Parameters. Record additional changes to operational setpoints and the time of the change in Attachment 9: Comments Section. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.	(OD)		
CDF-S13-038	CDF	\$13	Concentration 1	Instr.	Once the Virus Filtration product vessel is confirmed empty, Close manual valve H on the VF Product outlet. Ensure the following valve states:	D, E, F, G, H, I A, B, C	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-039	CDF	\$13	Concentration 1	Instr.	Continue Concentration 1 until the Target product weight in the Retentate Vessel after Concentration 1 (Step 12.5) is achieved. Once the Retentate Vessel reaches the weight in Step 12.5: Stop the pump on the UF/DF skid. Open the retentate valves (XV301 and XV302) completely. Close the permeate valve (XV201).	At the end of Concentration 1, ensure the following: A, D, E, F, G, H*, I*	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-040	CDF	\$13	Concentration 1	Instr.	Open		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-041	CDF	\$13	Concentration 1	Instr.	* NOTE: Valves H and I will remain closed for the rest of the UF/DF process.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-042	CDF	\$13	Concentration 1	Instr.	Record the volume of permeate from the permeate totalizer at the end of Concentration 1.	Permeate Volume	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S13-043	CDF	\$13	Concentration 1	Instr.	Record the net weight of the product at the end of Concentration 1 from the retentate scale and scale information.	Net Weight of Product after Concentration 1 Scale ID Cal. Due Date	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S14-001	CDF	\$14	Diafiltration	Instr.	Diafiltration (DF) Line Priming. Prime the Charge/DF Line with Diafiltration buffer (8012555) by performing the following: First, ensure valves A, D, E, F, and G are closed. Open bleed valve B-1. Ensure the tubing is directed to waste. Open valves E and F to prime the Charge/DF Line. If necessary, use a pump to prime the Charge/DF Line. Once no more air is observed exiting bleed valve B-1, close valve E, then close bleed valve B-1. Close valve F. Ensure that valves B and C are open. NOTE: Ensure no Diafiltration (DF) buffer enters the Retentate Vessel during the prime.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-002	CDF	\$14	Diafiltration	Instr.	Click on the "Add Recipe" button in Recipe Control screen. Find and Select DF1244_Diafiltration from the list. Record the Diafiltration batch ID and start time from the	Batch ID Start Time At the user prompt, enter the following values. Process Step	SMPL: Conditional Continued Diafiltration (block stack)	Cove

					system overview screen. The batch ID should follow this format: MPR Part Number_Diafiltration_Product Lot Number_Processing Date.			
CDF-S14-003	CDF	\$14	Diafiltration	Instr.	Input Title	Set Points Target Permeate Volume after Diafiltration Diafiltration_Perm_Vol Step 12.6 Trans-Membrane Pressure (psid) Diafiltration_TMP Cross Flow (L/min/ft2) Diafiltration_Crossflow	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-004	CDF	\$14	Diafiltration	Instr.	The TMP can be set to ≤ 35 psid as needed. If the TMP is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new value in Attachment 9: Comments Section.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-005	CDF	\$14	Diafiltration	Instr.	Ensure that valves E and F remain closed, then open valve A to prepare for Diafiltration. Push START to initiate the Diafiltration recipe		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-006	CDF	\$14	Diafiltration	Instr.	Diafiltration: Once the skid pump starts: Open valve E, then slowly open valve F to allow diafiltration buffer to enter the Retentate Vessel. Adjust valve F or the buffer pump speed to control the flow of DF buffer to the Retentate Vessel to maintain the weight from Step 13.30. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. Prior to the end of diafiltration, obtain samples from the retentate and permeate. Record the appropriate readings in Step 14.10. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.	Maximum TMP (psid) Feed Flow (L/min/ft2) Feed Pressure (psig) UV (OD)	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-007	CDF	\$14	Diafiltration	Instr.	At the completion of the Diafiltration recipe once the skid pump stops, close valves E and F.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-008	CDF	\$14	Diafiltration	Instr.	Record the permeate volume from the permeate totalizer after Diafiltration and ensure it is within the range calculated in Step 12.6.	Permeate Volume	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-009	CDF	\$14	Diafiltration	Instr.	Record the net weight of the product at the end of Diafiltration from the retentate scale and scale information.	Net Weight of Product after Diafiltration Scale ID Cal. Due Date	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-010	CDF	\$14	Diafiltration	Instr.	Calculate the product volume in the Retentate Vessel after Diafiltration.	Net Weight of Product after Diafiltration (Step 14.8) Density of the product after Diafiltration Product Volume in the Retentate Vessel after Diafiltration kg/L	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-011	CDF	\$14	Diafiltration	Instr.	Collect a sample from the permeate and retentate lines for pH, conductivity, and temperature measurement. Record the results below. NOTE: If the Retentate and Permeate pH values are not within range, contact a	Retentate Values Conductivity mS/cm Temperature (15 – 25°C) Permeate Values Conductivity mS/cm Temperature (15 – 25°C)	SMPL: Conditional Continued Diafiltration (block stack)	Cove

					supervisor prior to proceeding and document actions taken in Attachment 9: Comments Section.			
CDF-S14-012	CDF	\$14	Diafiltration	Instr.	Proceed based on the Step 14.10 sample results:	Check One The pH of the Step 14.10 retentate samples are: Actions to take In-range	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-013	CDF	\$14	Diafiltration	Instr.	Stop the transfer phase on the PCS HMI for the Diafiltration buffer (PN: 8012555. Close valves A and F. N/A Steps 14.12 through 14.22, then go to Section 15.	Out-of-range	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-014	CDF	\$14	Diafiltration	Instr.	Ensure Valves D, E, F, and G are closed. Ensure Valves A, B and C are opened. Go to the next step.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-015	CDF	\$14	Diafiltration	Instr.	Contact area supervisor to determine additional volume to be used for continued Diafiltration. NOTE: Additional Volume = Target Permeate Volume after Continued Diafiltration NOTE: There is no NOR maximum volume for Diafiltration.	Additional Volume Supervisor Contacted	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-016	CDF	\$14	Diafiltration	Instr.	Click on the "Add Recipe" button in Recipe Control screen. Find and Select DF1244_Diafiltration from the list. Record the Diafiltration batch ID and start time from the system overview screen. The batch ID should follow this format: MPR Part Number_Diafiltration2_Product Lot Number_Processing Date.	Batch ID Start Time At the user prompt, insert the following values. Process Step	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-017	CDF	\$14	Diafiltration	Instr.	Input Title	Set Points Target Permeate Volume after Diafiltration Diafiltration_Perm_Vol Step 14.12 Trans-Membrane Pressure (psid) Diafiltration_TMP Cross Flow (L/min/ft2) Diafiltration_Crossflow The TMP can be set to ≤ 35 psid as needed. If the value is adjusted, record the value in Attachment 9: Comments Section. Ensure the feed pressure is ≤ 60 psig. The cross flow can be adjusted within the range of 0.3 – 0.7 L/min/ft2, as needed. If the value is adjusted, record the new the value in Attachment 9: Comments Section	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-018	CDF	\$14	Diafiltration	Instr.	Ensure that valves E and F remain closed, then open valve A to prepare for Diafiltration. Push START to initiate the Diafiltration recipe.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-019	CDF	\$14	Diafiltration	Instr.	Additional Diafiltration: Once the skid pump starts: Open valve E, then slowly open valve F to allow the diafiltration buffer to enter the Retentate Vessel. Adjust valve F or the buffer pump speed to control the flow of DF buffer to the Retentate Vessel to maintain the weight from Step 13.30. Record the process data in Attachment 2: Operation Parameters approximately every 20 minutes. Prior to the end of Diafiltration, obtain samples from the retentate and permeate. Record the appropriate readings in Step 14.21. NOTE: If any of the maximum values below are	Maximum TMP (psid) Feed Flow (L/min/ft2) Feed Pressure (psig) UV (OD)	SMPL: Conditional Continued Diafiltration (block stack)	Cove

					exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.			
CDF-S14-020	CDF	\$14	Diafiltration	Instr.	At the completion of the Diafiltration recipe once the skid pumps stops, close valves E and F.		SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-021	CDF	\$14	Diafiltration	Instr.	Record the permeate volume after continued diafiltration and ensure it is less than or equal to the volume calculated in Step 14.12.	Permeate Volume	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-022	CDF	\$14	Diafiltration	Instr.	Record the net weight of the product at the end of diafiltration from the retentate scale and scale information.	Net Weight of Product after Diafiltration Scale ID Cal. Due Date	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-023	CDF	\$14	Diafiltration	Instr.	Calculate the product volume in the Retentate Vessel after the additional Diafiltration.	Net Weight of Product after Diafiltration (Step 14.19) Estimated Density of the Product after Diafiltration Product Volume in the Retentate Vessel after Diafiltration kg/L	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-024	CDF	\$14	Diafiltration	Instr.	Collect a sample from the permeate and retentate lines for pH, conductivity, and temperature measurement. Record the results below.	Retentate Values Conductivity mS/cm Temperature (15 – 25°C) Permeate Values Conductivity mS/cm Temperature (15 – 25°C)	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-025	CDF	\$14	Diafiltration	Instr.	Proceed based on the Step 14.21 sample results:	Check One The pH of the Step 14.21 samples are: Actions to take In-range	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-026	CDF	\$14	Diafiltration	Instr.	Stop the transfer phase on the PCS HMI for the Diafiltration buffer. Close valves E and F. Go to Section 15.	Out-of-range	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S14-027	CDF	\$14	Diafiltration	Instr.	Contact an area supervisor and document actions take in Attachment 9: Comments Section.	Name of supervisor contacted	SMPL: Conditional Continued Diafiltration (block stack)	Cove
CDF-S15-001	CDF	\$15	Concentration 2	Instr.	Prior to starting Concentration 2: Reset the permeate totalizer. Ensure the valve states below.	A, D, E, F, G	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-002	CDF	\$15	Concentration 2	Instr.	To begin Concentration 2, click on the “Semi-Automatic” mode on the UF/DF Skid screen and input the following initial parameters in the order listed. NOTE: Drain valves XV102, XV103, XV202 should not opened at any time during processing. The feed pressure can be adjusted to as needed to a maximum AL feed pressure $\leq$ 60 psig. Record changes to transmembrane pressure and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section. The TMP can be set to $\leq$ 35 psid as needed. If the value is adjusted, notify supervisor if TMP is $\geq$ 35 psid record the value in Attachment 9: Comments Section. Ensure the feed pressure is $\leq$ 60 psig. Feed pressure NOR is $\leq$ 60 psig.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-003	CDF	\$15	Concentration 2	Instr.	Push START to initiate the Concentration 2 process. Record the start time as the time the pump turns on.	Start Time	SMPL: Run Skid Recipe (block stack)	Cove

CDF-S15-004	CDF	\$15	Concentration 2	Instr.	Concentration 2: The crossflow and transmembrane pressure (TMP) setpoint(s) may be adjusted from their initial values as necessary as long as they do not exceed the maximum values below. Record the adjusted values, the time of adjustment, and the step number in Attachment 2: Operation Parameters. Record additional changes to operational setpoints and the time of the change in Attachment 9: Comments Section. Record the process data and step number in Attachment 2: Operation Parameters approximately every 20 minutes. Concentration 2 is complete when the retentate weight reaches the Target net weight calculated in Step 12.9. Proceed to the next step for instructions to end Concentration 2. NOTE: If any of the maximum values below are exceeded, stop process immediately, contact an area supervisor, and document actions taken in Attachment 9: Comments Section.	Maximum TMP (psid) Feed Flow (L/min/ft2) Feed Pressure (psig) UV (OD)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-005	CDF	\$15	Concentration 2	Note	When the target weight of the Retentate Vessel after Concentration 2 is reached (Step 12.9): Stop the pump. Ensure the valve states below:	A, D, F, G Permeate (XV201) B, C, Retentate (XV301 and XV302)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-006	CDF	\$15	Concentration 2	Instr.	Record the end time, retentate weight and the actual volume of permeate from the permeate totalizer after Concentration 2. * NOTE: Ensure the Retentate Weight is within the range calculated in Step 12.9	Concentration 2 End Time Retentate Weight * Permeate Volume	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-007	CDF	\$15	Concentration 2	Instr.	Recirculation: To recirculate the concentrated product, click on the "Semi-Automatic" mode on the UF/DF Skid screen and input the following initial parameters in the order listed: NOTE: Drain Valves XV102, XV103, XV202 should not be opened at any time during processing. The feed pressure can be adjusted as needed to maintain a target of 25 psig and a maximum feed pressure ≤ 60 psig. Record changes to TMP and/or cross flow setpoint(s) in Attachment 2: Operation Parameters. Record any additional changes in Attachment 9: Comments Section The cross flow can be adjusted, as long as crossflow is ≤ 0.7 L/min/ft2, to obtain the target feed pressure. Do not exceed 60 psig for the feed pressure. If the value is adjusted, record the new value in Attachment 9: Comments Section.		SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-008	CDF	\$15	Concentration 2	Instr.	Recirculate for 5 - 120 minutes (max NOR for recirculation is 120 mins). Record the start and end time of recirculation below. Once recirculation is complete: Stop the pump. Close valves B and C. Close the Feed Valve (XV101) on the HMI.	Recirculation start time Recirculation end time Duration of recirculation (5 - 120 min)	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S15-009	CDF	\$15	Concentration 2	Instr.	Ensure the Retentate Vessel is in the same configuration as it was when its tare weight after attachments was obtained in Step 11.12.		SMPL: Run Skid Recipe (block stack)	Cove

CDF-S15-010	CDF	\$15	Concentration 2	Instr.	Record the weight of the Retentate Vessel after Recirculation.	Retentate Weight after Recirculation Scale ID Cal. Due Date	SMPL: Run Skid Recipe (block stack)	Cove
CDF-S16-001	CDF	\$16	Recovery	Instr.	Calculate the Recovery Product volume range.	Total grams of Virus Filtration Product after sampling (Step 9.10) Recovery Concentration Range Recovery Product Volume Range L (Target) L (Max)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-002	CDF	\$16	Recovery	Instr.	Calculate the Recovery Product weight range.	Recovery Product Volume Range (Step 16.1) Density of the Product at Recovery Recovery Product Net Weight Range L (Target) 1.065 kg/L kg (Target) L (Max) 1.065 kg/L kg (Max)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-003	CDF	\$16	Recovery	Instr.	Calculate the amount of Recovery buffer to add.	Recovery Product Net Weight range (Step 16.2) Retentate Weight after Recirculation (Step 15.10) Weight of Recovery buffer to add kg (Target) kg (Target) kg (Max) kg (Max)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-004	CDF	\$16	Recovery	Instr.	Calculate the volume of Recovery buffer to add.	Target Weight of Recovery buffer to add (Step 16.3) Density of the Recovery buffer Volume of Recovery buffer to add 1.008 kg/L	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-005	CDF	\$16	Recovery	Instr.	Prior to Recovery, ensure the valve states listed below:	A, B, F, G, Feed (XV101), Permeate (XV201) C, D, E, Retentate (XV301 and XV302)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-006	CDF	\$16	Recovery	Instr.	Choose below whether Recovery will be performed manually or by recipe control:	Check One Condition Action to take Recovery will be run manually N/A Step 16.7 to Step 16.11. Go to Step 16.12. Recovery will be run by recipe control Go to the next step. Recovery by Recipe Control (Steps 16.7 – 16.11) Click on the “Add Recipe” button on the Recipe Control screen. Select DF1244_RECOVERY_2 from the list. Record the recovery batch ID and start time from the system overview screen. The batch ID should follow this format: MPR part Number_Recovery_Product Batch Number_Processing Date Batch ID Start Time At the user prompt, insert the following values. Process Step	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-007	CDF	\$16	Recovery	Instr.	Input Title	Set Points Retentate Volume Recovery_Ret_Vol Step 16.4 Max Feed Pressure Feed_Pressure_Max 60 psig a Target feed pressure is 15 psig. Push START to initiate the Recovery process.	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-008	CDF	\$16	Recovery	Instr.	Once the pump starts, slowly open manual valve G to allow Recovery buffer to flow into the Retentate Vessel and to maintain a feed flow of less than or equal to 0.7	Supervisor Contacted	SMPL: Recovery Calculations & Execution (block stack)	Cove

					L/min/ft2. If the feed pressure exceeds 60 psig, stop the process, contact area management, and document actions taken in Attachment 9: Comments Section. Otherwise mark the box below as "NA"			
CDF-S16-009	CDF	\$16	Recovery	Instr.	At the completion of Recovery when the skid pump stops, immediately close valves C and G to stop the flow of Recovery buffer into the Retentate Vessel. Ensure the remaining manual valves on the UF/DF system are closed per the table below, then N/A Steps 16.12 – 16.13 and proceed to Step 16.14.	A, B, C, D, E, F, G Manual Recovery (Steps 16.12 – 16.13) Prior to the start of Recovery, verify the valve states below on the UF/DF System: A, B, F, G, Feed (XV101), Permeate (XV201) C, D, E, Retentate (XV301 and XV302)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-010	CDF	\$16	Recovery	Instr.	Open Feed valve XV101 on the skid HMI, then slowly open manual valve G to allow Recovery buffer to flow into the Retentate Vessel and to maintain a feed flow of less than or equal to NOR max of 0.7 L/min/ft2. Monitor the weight of the Retentate Vessel until the target Recovery Product Net Weight (Step 16.2) is achieved, then immediately close valves C and G to stop the flow of Recovery buffer into the Retentate Vessel. Ensure the remaining manual valves on the UF/DF system are closed per the table below.	A, B, C, D, E, F, G	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-011	CDF	\$16	Recovery	Instr.	Obtain the weight of the product after Recovery (with attachments). Record the scale/balance equipment ID and calibration due date.	Weight of product after Recovery (with attachments) Scale/Balance ID and Cal Due Date Due Date	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-012	CDF	\$16	Recovery	Instr.	Calculate the amount of Recovery buffer 8012555 used for the Recovery flush.	Retentate Vessel Weight after Recovery (with attachments) (Step 16.14) Retentate Weight after Recirculation (Step 15.10) Amount of Recovery buffer 8012555 used for Recovery flush	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-013	CDF	\$16	Recovery	Instr.	Close all the valves on the Retentate Vessel. Disconnect the following parts: The Charge/Diafiltration Line with sight glass, elbow, and tee connected to valve A. NOTE: Keep valve A connected to the tank. The Retentate Line with sight glass and tee connected to valve C. NOTE: Keep valve C connected to the tank. The Recovery and Feed Lines including the reducer and tee connected to the valve B tank outlet. NOTE: This will include disconnecting valve D.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-014	CDF	\$16	Recovery	Instr.	Place end caps on valves A, B, and C. The Retentate Vessel should be in the configuration per the diagram below.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-015	CDF	\$16	Recovery	Instr.	Obtain a gross weight of the Retentate Vessel, with no attachments, after Recovery. Ensure that the retentate vessel is the same configuration as in Step 16.17. If necessary, remove the Retentate Vessel from the scale and tare (zero) the scale again before obtaining the gross weight. Record the scale/balance equipment ID and calibration due date.	Gross Weight of Retentate Vessel after Recovery (no attachments) Scale/Balance ID and Cal Due Date Due Date	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-016	CDF	\$16	Recovery	Instr.	Obtain a mixing bag for the transfer of the Pre-Formulated Intermediate (PFI) product. Record the part	Dilution Vessel Info	SMPL: Recovery Calculations & Execution (block stack)	Cove

					number, batch number, and expiration date for the bag. Ensure the vessel is labeled with "AZD0543", "Pre-Formulated Intermediate Dilution Vessel", batch number, 8012475, and initials/date.			
CDF-S16-017	CDF	\$16	Recovery	Instr.	Obtain the tare weight for the Dilution Vessel. Record the equipment ID and calibration due date for the scale and the tare weight of the bag. Check if the Mixer Drive is attached when the tare weight is obtained.	Scale/Balance ID Scale/Balance Calibration Due Date Tare Weight of Dilution Vessel before attachment Mixer Drive attached at Tare Weight? (check one) ☐ Yes ☐ No	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-018	CDF	\$16	Recovery	Instr.	Obtain tubing/hose(s) for the transfer of the product to the Dilution Vessel. Record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. Record the hose cleaning cycle number and expiration date, if applicable.	Tubing Info, if applicable Hose Info, if applicable Hose Cleaning Cycle No. Hose Cleaning Exp. Date * Autoclave information is not required if tubing is gamma irradiated.	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-019	CDF	\$16	Recovery	Note	If the transfer line is tubing, thread the tubing through the peristaltic pump following SOP-0080839.	Turn off the agitator on the Retentate Vessel.	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-020	CDF	\$16	Recovery	Instr.	Connect the Recovery line from the outlet of Retentate Vessel elbow to the inlet of the Dilution Vessel. NOTE: Ensure the Dilution Vessel bottom outlet clamp is closed and clamped as close to the bag as possible.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-021	CDF	\$16	Recovery	Instr.	Once the Dilution Vessel is connected to the outlet of the Retentate Vessel, open the outlet of the Retentate Vessel at valve B. Pump the entire contents of the Retentate Vessel into the Dilution Vessel and record the end time of transfer below.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-022	CDF	\$16	Recovery	Instr.	Transfer End Time		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-023	CDF	\$16	Recovery	Instr.	Begin to mix the product by agitation per SOP-0107097. Ensure the agitation is set to mix the product while minimize foaming. Mix for at least 20 minutes. Once a steady agitation rate has been achieved, record the mixer ID, agitation setting, mixing start time, mixing end time and the mixing duration below.	Mixer ID Agitation Setting Mixing Start Time Mixing End Time Mixing Duration (≥20 mins)	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-024	CDF	\$16	Recovery	Instr.	Remove product from the Dilution Vessel via a sample port for STAT protein determination (-20) per Attachment 3: Sampling Record. Follow procedures in SOP-0107097.	Sampling Date and Time Date: Time:	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-025	CDF	\$16	Recovery	Instr.	Aliquot the sample (sample designation: 8012475-20) per Attachment 3: Sampling Record. If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample with 8012475-20, per SOP-0107056.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-026	CDF	\$16	Recovery	Instr.	Submit 8012475-20 according to SOP-0080295. Ensure the sampling time and location are updated in DLIMS.		SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-027	CDF	\$16	Recovery	Instr.	Remove from the Dilution Vessel the tubing/hosing that was used for transfer. NOTE: Ensure the vessel is in the		SMPL: Recovery Calculations & Execution (block stack)	Cove

					same configuration as it was when the tare weight before attachment was obtained in Step 16.20.			
CDF-S16-028	CDF	\$16	Recovery	Instr.	Record the gross weight of the Dilution Vessel after transfer. Record the floor scale/balance ID number and calibration due date.	Gross weight of the Dilution Vessel Scale/Balance ID Scale/Balance Calibration Due Date	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-029	CDF	\$16	Recovery	Instr.	Calculate the net weight of the Recovery Product after transfer.	Gross weight of the Dilution Vessel (Step 16.31) Tare weight of Dilution Vessel before attachment (Step 16.20) Net weight of Recovery Product after Transfer	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-030	CDF	\$16	Recovery	Instr.	Calculate the volume of the Recovery Product after Transfer.	Net weight of Recovery Product after Transfer (Step 16.32) Density of the product at 190 g/L Volume of Recovery Product after Transfer kg/L	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-031	CDF	\$16	Recovery	Instr.	Record the protein concentration and DLIMS Project and Sample ID of the Recovery Product (8012475-20). Based on the 8012475-20 concentration result, indicate below if the product is within the Pre-Formulated Intermediate (PFI) concentration range of 171 – 189 g/L or if it requires dilution.	Concentration of the Recovery Product DLIMS Project ID DLIMS Sample ID Check One Concentration Action > g/L Dilution is required: Go to Section 17. – g/L	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-032	CDF	\$16	Recovery	Instr.	Dilution is not required: N/A Steps 17.1 - 17.16, then go to Step 17.17.	< g/L	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S16-033	CDF	\$16	Recovery	Instr.	Contact a supervisor and document actions taken in Attachment 9: Comments Section.	Supervisor Contacted, if applicable	SMPL: Recovery Calculations & Execution (block stack)	Cove
CDF-S17-001	CDF	\$17	Dilution	Instr.	Calculate the total grams of product in the Retentate Vessel.	Volume of Recovery Product after Transfer (Step 16.33) Concentration of the Recovery Product (Step 16.34) Total grams of the Recovery Product	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-002	CDF	\$17	Dilution	Instr.	Calculate the AZD0543 Dilution Product Volume Range.	Total grams of the Recovery Product (Step 17.1) PFI Concentration Range AZD0543 Dilution Product Volume Range g/L (max) L (Min) 180 g/L (target) L (Target) g/L (min) L (Max)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-003	CDF	\$17	Dilution	Instr.	Calculate the AZD0543 Dilution Product Net Weight Range.	AZD0543 Dilution Product Volume Range (Step 17.2) Density of the Pre-Formulated Intermediate (PFI) AZD0543 PFI Net Weight Range L (Min) 1.062 kg/L kg (Min) L (Target) 1.062 kg/L kg (Target) L (Max) 1.062 kg/L kg (Max)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-004	CDF	\$17	Dilution	Instr.	Calculate the weight of Dilution Buffer to Add.	AZD0543 PFI Net Weight Range (Step 17.3) Net weight of the Recovery Product after Transfer (Step 16.32) Weight of Dilution Buffer to Add kg (Min) kg (Min) kg (Target) kg (Target) kg (Max) kg (Max) Convert the Weight of Dilution Buffer to Add to volume. Weight of Dilution Buffer to Add (Step 17.4) Density of the Dilution	SMPL: Dilution Decision & Calculations (block stack)	Cove

						Buffer (PN: 8012555) Volume of Dilution Buffer to Add kg (Min) 1.008 kg/L L (Min) kg (Target) 1.008 kg/L L (Target) kg (Max) 1.008 kg/L L (Max)		
CDF-S17-005	CDF	\$17	Dilution	Instr.	Check the box for the method for how the dilution buffer will be transferred.	Check One Dilution Method Action	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-006	CDF	\$17	Dilution	Note	If using a pipette, grad cylinder, carboy, or bottle		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-007	CDF	\$17	Dilution	Instr.	Proceed to the next step.		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-008	CDF	\$17	Dilution	Note	If connecting the dilution buffer directly to the Dilution Vessel	N/A Steps 17.7 – 17.9. Go to Step 17.10.	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-009	CDF	\$17	Dilution	Instr.	Obtain a vessel for the transfer of the dilution buffer. Label the vessel as “Intermediate Vessel”, and initial and date. NOTE: N/A any unused spaces.	Qty Carboy Grad. Cylinder Bottle Sterile Pipette Sterile Pipette Tubing	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-010	CDF	\$17	Dilution	Instr.	Check the box for the method of addition. If adding using a pump, record the pump ID, if applicable.	Pump ID, if applicable Measurement Method (range reference listed) Volumetrically Step 17.5 Weight Step 17.4	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-011	CDF	\$17	Dilution	Instr.	Transfer the dilution buffer (8012555) to the Intermediate vessel.		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-012	CDF	\$17	Dilution	Note	If transferring using a pump, connect either the dilution buffer (8012555) or the Intermediate Vessel to the inlet of the PFI Dilution vessel. Thread tubing through a peristaltic pump. Prime the tubing. Otherwise, mark this step as “NA.”	Tubing Information Part Number Batch Number Expiration Date * Autoclave information is not required if tubing is gamma irradiated. Tare (zero) the scale holding the Dilution Vessel.	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-013	CDF	\$17	Dilution	Instr.	Add dilution buffer to the Dilution Vessel until the target Weight of Dilution Buffer to Add (Step 17.4) is reached. Once target weight is reached, turn off the transfer pump. Record weight after dilution.	Net Weight of dilution buffer 8012555 added Scale/Balance ID Cal. Due Date	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-014	CDF	\$17	Dilution	Instr.	Disconnect the buffer line from the Dilution Vessel, if applicable. Remove the dilution vessel from the scale and tare (zero) the scale. Obtain the gross weight of the Dilution Vessel after dilution. NOTE: Ensure the Dilution Vessel is in the same configuration as it was when its tare weight was obtained in Step 16.20.	Gross weight of Dilution Vessel, with attachments removed Scale/Balance ID Calibration Due Date	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-015	CDF	\$17	Dilution	Instr.	Calculate the net weight of PFI.	Gross weight of Dilution Vessel, with attachments removed (Step 17.13) Tare weight of Dilution Vessel before attachment (Step 16.20) PFI Net Weight	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-016	CDF	\$17	Dilution	Note	If the PFI Net Weight is above the max net weight range calculated in Step 17.3, contact a supervisor and document actions taken in Attachment 9: Comments Section. Otherwise, mark this step as “NA”.	Supervisor Contacted	SMPL: Dilution Decision & Calculations (block stack)	Cove

CDF-S17-017	CDF	\$17	Dilution	Instr.	Calculate the volume of PFI after dilution and before sampling.	PFI Net Weight (Step 17.14) Density of the PFI PFI Volume before sampling 1.062 kg/L	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-018	CDF	\$17	Dilution	Instr.	Begin to mix the product per SOP-0107097 by agitation. If product is mixed by recirculation, complete Attachment 7: Product Recirculation Work Sheet. Mix the product for at least 20 minutes by agitation. Adjust agitation setting to minimize foaming, as applicable. NOTE: Agitation may already be on.	Mixer/Stir Plate ID, if applicable Agitation Setting Mixing Start Time Mixing End Time Duration of Mixing (≥ 20 minutes)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-019	CDF	\$17	Dilution	Instr.	Select one of the options below.	Check One Condition Actions to take	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-020	CDF	\$17	Dilution	Note	If there was no dilution		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-021	CDF	\$17	Dilution	Instr.	Mark Steps 17.19 – 17.23 as “NA.” Mark sample (8012475 – 30) in Attachment 3: Sampling Record as “N/A”. Proceed to Step 17.24.		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-022	CDF	\$17	Dilution	Note	If there was dilution		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-023	CDF	\$17	Dilution	Instr.	Proceed to the next step		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-024	CDF	\$17	Dilution	Instr.	Remove product from the product collection vessel for (-30) samples per Attachment 3: Sampling Record. Sample per SOP-0107097. NOTE: Ensure the sample line is flushed prior to taking the sample.	Sampling Time and Date Time Date	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-025	CDF	\$17	Dilution	Instr.	Aliquot the sample (sample designation: 8012475-30) per Attachment 3: Sampling Record If applicable, aliquot non-routine samples per Attachment 4: Non-Routine Sampling Record. Label the sample per SOP-0107056		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-026	CDF	\$17	Dilution	Instr.	Submit the sample in DLIMS per SOP-0080295. Ensure the sampling time and location are updated in DLIMS.		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-027	CDF	\$17	Dilution	Instr.	Record the results from sample designation, 8012475–30. If the results are not in range, contact supervisor and document actions taken in Attachment 9: Comments Section.	PFI concentration (– g/L)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-028	CDF	\$17	Dilution	Instr.	Once the Pre-Formulated Intermediate has confirmed to be in range, begin to mix the product per SOP-0107097 by agitation. If using a stir plate, record the equipment ID for the stir plate used. NOTE: Agitation may already be on.	Mixer/Stir Plate ID Agitation Setting Mixing Start Time Mixing End Time Mixing Duration (≥ 15 minutes)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-029	CDF	\$17	Dilution	Instr.	Remove product from the product collection vessel for (-40) samples per Attachment 3: Sampling Record. Follow procedures in SOP-0107097 for mixing bags.	Sampling Time and Date Time Date	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-030	CDF	\$17	Dilution	Instr.	Aliquot the sample (sample designation: 8012475-40) per Attachment 3: Sampling Record. If applicable, aliquot		SMPL: Dilution Decision & Calculations (block stack)	Cove

					non-routine samples per Attachment 4: Non-Routine Sampling Record.			
CDF-S17-031	CDF	\$17	Dilution	Instr.	Label with 8012475 and additional information per SOP-0107056 Submit the sample in DLIMS per SOP-0080295. NOTE: Ensure the sampling time and location are updated in DLIMS.		SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-032	CDF	\$17	Dilution	Instr.	Aliquot product into a 50 ml conical and measure the pH, conductivity, and temperature. NOTE: If the pH value is not in range, contact a supervisor and document actions taken in Attachment 9: Comments Section.	Conductivity mS/cm Temperature (15 – 25 °C)	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-033	CDF	\$17	Dilution	Instr.	Record the gross weight of the Dilution Vessel after sampling. NOTE: Ensure the vessel is in the same configuration as it was when the tare weight was obtained in Step 16.20.	Dilution Vessel Gross Weight after sampling	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-034	CDF	\$17	Dilution	Instr.	Calculate the net weight of the Pre-Formulated Intermediate (PFI) after sampling.	Dilution Vessel Gross Weight after sampling (Step 17.30) Tare weight of Dilution Vessel (Step 16.20) PFI Net Weight after sampling	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S17-035	CDF	\$17	Dilution	Instr.	Calculate the volume of the PFI after sampling.	PFI Net Weight after sampling (Step 17.31) Density of the PFI PFI Volume after sampling 1.062 kg/L	SMPL: Dilution Decision & Calculations (block stack)	Cove
CDF-S18-001	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Obtain a mixing bag or bottle for the filtration and storage of the PFI. Record the bag or bottle information below and ensure the vessel is labeled with "AZD0543", "Pre-Formulated Intermediate", batch number, part number, and initials/date. NOTE: A mixing bag is preferred to use for storage.	Bag information (if applicable) Bag Part No. Bag Batch No. Bag Exp. Date Glass Bottle Information (if applicable) Glass Bottle Description Glass Bottle Autoclave Cycle No. Glass Bottle Autoclave Exp. Date	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-002	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Obtain a tare weight of the PFI storage vessel. Record the equipment ID and calibration due date for the scale.	Scale/Balance ID Scale/Balance Calibration Due Date Tare Weight of Storage Vessel	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-003	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Obtain an SHC filter per Section 4 and record the filter information below. NOTE: The bolded filter part number(s) in Section 4 require autoclaving.	Pre-Formulated Intermediate Product Filter Information Part Number	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-004	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Obtain tubing/hose(s) for the transfer of the product to the storage vessel. Record the tubing part number, batch number, expiration date, autoclave cycle number, and autoclave expiration date. Record the hose cleaning cycle number and expiration date, if applicable.	Tubing Information, if applicable Hose Information, if applicable Hose Cleaning Cycle No. Hose Cleaning Exp. Date * Autoclave information is not required if tubing is gamma irradiated.	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-005	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Attach the outlet of the PFI product filter to the inlet of the PFI storage vessel. Connect the transfer line between the outlet of the Dilution Vessel and the inlet of the PFI product filter. If the transfer line is tubing, thread the tubing through the peristaltic pump following SOP-0080839. Pump the entire contents of the Dilution Vessel into the PFI storage vessel and record the end time of transfer below.		SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove

CDF-S18-006	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Transfer End Time		SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-007	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Drain the hold-up of the PFI product filter and transfer line into the PFI storage vessel. Remove the transfer tubing from the PFI storage vessel and remove the PFI product filter(s).		SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-008	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Obtain a gross weight of the PFI storage vessel after the transfer. Record the scale/balance equipment ID and calibration due date. NOTE: Ensure the storage vessel is in the same configuration as when the tare weight was obtained in Step 18.2	Gross Weight of PFI Storage Vessel after transfer Scale/Balance ID Scale/Balance Calibration Due Date	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-009	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Calculate the net weight of the PFI after sampling and transfer	Gross Weight of PFI Storage Vessel after transfer (Step 18.7) Tare Weight of Storage Vessel (Step 18.2) Net Weight of the PFI after sampling and transfer	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-010	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Calculate the volume of the PFI after sampling and transfer.	Net Weight of the PFI after sampling and transfer (Step 18.8) Density of the Product after Dilution Volume of the PFI after sampling and transfer 1.062 kg/L	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-011	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Record the net weight of the PFI (Step 18.8), the tare weight of the PFI storage vessel (Step 18.2), and the gross weight of the PFI storage vessel after transfer (Step 18.7) on the storage vessel label.		SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-012	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Store the product and document storage information below. For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage. For product stored at 15 – 25°C, the storage start time is the time when final sampling occurred in Step 17.24.		SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S18-013	CDF	\$18	Filtration and Storage of the Pre-Formulated Intermediate	Instr.	Record the following values in Attachment 5: Hold Times.	MPR Name Name in Attachment Step No. Start Time/Date of Product Charge Virus Filtered Product Storage End (Time/Date) Step 13.14	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	Cove
CDF-S19-001	CDF	\$19	Post-Use WFI Flush	Instr.	Set up per SOP-0107111 using one of the options below. For WFI collected directly from the WFI Hold Tank (TK-1458) Verify the SIP of the WFI Hold Tank was completed and the tank status is "Clean" and that the WFI port has been flushed per SOP-0080303. Mark the bag information as "N/A." For WFI collected into a bag If the same WFI vessel from Step 8.1 is used: Transcribe from Step 8.1 the WFI Valve/Tank ID, WFI Charge Time/Date and WFI Expiration Date/Time below. N/A the bag information boxes and N/A the SAP consumption box. Record the WFI collection vessel information. Verify that the WFI port has been flushed per SOP-0080185. Charge	Bag, if applicable WFI Valve ID/Tank ID WFI Charge Date/Time Date: Time: WFI Expiration Date/Time * Date: Time:	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove

					into the vessel and record the WFI Charge Date/Time and Expiration.			
CDF-S19-002	CDF	\$19	Post-Use WFI Flush	Instr.	* The charged WFI Expiration Date/Time is 24 hours from the WFI Charge Date/Time.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-003	CDF	\$19	Post-Use WFI Flush	Instr.	Confirm that the conductivity meter has been standardized per SOP-0080297.	Meter ID	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-004	CDF	\$19	Post-Use WFI Flush	Instr.	Ensure that the Ultrafiltration (UF) system is set up according to SOP-0107111.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-005	CDF	\$19	Post-Use WFI Flush	Instr.	On the UF/DF Skid HMI, enter the total cassette surface area in m2 (Step 7.2): Click on the membrane Area Setpoint input box. Enter surface area in m2 (Step 7.2) and press ENTER. Click ACCEPT on the Membrane Area Change pop-up window.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-006	CDF	\$19	Post-Use WFI Flush	Instr.	Connect the WFI supply inlet to the UF/DF Skid feed line. Ensure that the retentate and permeate lines to the UF/DF Skid are directed to waste.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-007	CDF	\$19	Post-Use WFI Flush	Instr.	Download and run DF1244_Flush according to SOP-0107111 to flush the UF/DF Skid with WFI. Record the batch ID and start time. The batch ID should follow this format: MPR Part Number_CleanWFI_Batch Number_Processing Date	Batch ID Start Time At the user prompt, input the values per the table below. Processing Step	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-008	CDF	\$19	Post-Use WFI Flush	Instr.	Input Title	Entered Set Point Permeate Flush PERMEATE_FLUSH_VOL ≥ Step 8.7 Retentate Flush RET_FLUSH_VOL ≥ Step 8.7	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-009	CDF	\$19	Post-Use WFI Flush	Instr.	Go to the Recipe Control screen on the UF/DF Skid HMI, and push START to initiate the post-use WFI flush.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-010	CDF	\$19	Post-Use WFI Flush	Instr.	At the completion of the recipe, the status changes from "Running" to "Complete" and the pump stops. Continue to Section 20.		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-011	CDF	\$19	Post-Use WFI Flush	Instr.	Post-Processing Activities for the UF/DF Skid		SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-012	CDF	\$19	Post-Use WFI Flush	Instr.	Post-Processing Activities for the UF/DF Skid Check which action will be taken for the cassettes.	Check One Decision Actions to take The cassettes will not be re-used: Discard cassettes *	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-013	CDF	\$19	Post-Use WFI Flush	Instr.	N/A Step 20.2, then continue to Step 20.3 to clean and store the UF/DF skid. * NOTE: If the cassettes are to be discarded, NWP is not required.	The cassettes will be re-used: Clean, WFI Flush, perform NWP, and store the cassettes Continue to the next step.	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S19-014	CDF	\$19	Post-Use WFI Flush	Note	Verify the data for the UF/DF Skid to be used. From the logbook, record the logbook number, CIPDS batch	Logbook Number CIPDS Batch Number Action Date Sanitization	SMPL: Post-Processing: Cassette Re-use & Sanitization (block	Cove

					number and date when the actions were documented below.	NWP Storage	stack)	
CDF-S19-015	CDF	\$19	Post-Use WFI Flush	Instr.	Record the Sanitization and Storage Buffer Batch Number and amount used based on SOP-0107111 * Amount Used = Retentate Totalizer + Permeate Totalizer volume	Buffer Info Batch Number Amount Used (L) * Sanitization PN: A0013-D 0.5 N Sodium Hydroxide – DV Storage PN: A0014-D 0.1 N Sodium Hydroxide	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	Cove
CDF-S20-001	CDF	\$20	Batch Production Record Review	Fields		Production Initial/Date	SMPL: Instruction & Sign-off	Cove
CDF-S20-002	CDF	\$20	Batch Production Record Review	Note	If tanks were used, ensure that the SIP, CIP, and Pressure Test Reports for the product tank are attached to the MPR.		SMPL: Instruction & Sign-off	Cove
CDF-S20-003	CDF	\$20	Batch Production Record Review	Note	Verify that the PFI product net weight (from Step 18.8) is correctly inputted into SAP.		SMPL: Instruction & Sign-off	Cove
CDF-S20-004	CDF	\$20	Batch Production Record Review	Instr.	Review the Batch Production Record for completeness and submit for review.		SMPL: Instruction & Sign-off	Cove
CDF-S21-001	CDF	\$21	Attachment 1: Process Tubing/Hose Information	Fields		Hose information for the Retentate Vessel, if applicable Hose Serial Number Cleaning Exp. Date Feed Line Additional Feed Line, as needed Retentate Line Additional Retentate Line, as needed Permeate Line Additional Permeate Line, as Needed	SMPL: Transfer Tubing / Hosing Information	Cove
CDF-S21-002	CDF	\$21	Attachment 1: Process Tubing/Hose Information	Instr.	Charge/Diafiltration Line	Additional Charge/Diafiltration Line, if needed Additional Charge/Diafiltration Line, if needed Recovery Line Additional Recovery Line, if needed Witnessed/Date	SMPL: Transfer Tubing / Hosing Information	Cove
CDF-S21-003	CDF	\$21	Attachment 1: Process Tubing/Hose Information	Instr.	Attachment 1: Process Tubing/Hose Information (continued)	Tubing for the Retentate Vessel, if applicable Tubing Use Part Number Batch Number Expiration Date Feed Line Additional Feed Line, as needed Retentate Line Additional Retentate Line, as needed Permeate Line Additional Permeate Line, as Needed	SMPL: Transfer Tubing / Hosing Information	Cove
CDF-S21-004	CDF	\$21	Attachment 1: Process Tubing/Hose Information	Instr.	Charge/Diafiltration Line	Additional Charge/Diafiltration Line, if needed Peristaltic Pump Tubing Additional Peristaltic Pump Tubing, if needed Witnessed/Date	SMPL: Transfer Tubing / Hosing Information	Cove
CDF-S22-001	CDF	\$22	Attachment 2: Operation Parameters	Fields		Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove

						Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)		
CDF-S22-002	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-003	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-004	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-005	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-006	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-007	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed -	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove

						PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)		
CDF-S22-008	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-009	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-010	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-011	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove

CDF-S22-012	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-013	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-014	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-015	CDF	\$22	Attachment 2: Operation Parameters	Instr.	Attachment 2: Operation Parameters (continued)	Time PFeed (psig) @ PT-101 PRetentate (psig) @ PT-301 PPermeate (psig) @ PT-201 ΔP (psid) (= PFeed - PRetentate) Diafiltration Pump Settings, if applicable % Retentate Open (%) @ BPRV-301 Pump Speed (RPM) @ P-101 Crossflow Flux (L/min/ft2) TMP (psid) @ TMP-100 Permeate Flow Rate (L/min) @ FT-201 Permeate Flux (LMH) Total Permeate Volume (L) @ FQI-201 Permeate UV Reading (OD) @ AIT-201 Permeate Conductivity, offline (mS/cm) * Permeate pH, offline * Retentate Vessel Agitator Setting Product weight in Retentate Vessel (kg)	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S22-016	CDF	\$22	Attachment 2: Operation Parameters	Instr.	* = The conductivity meter used must match the meter used in Step 8.2, including temperature compensation during equilibration and sections forward. Otherwise, annotate in Attachment 9: Comments Section.		SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	Cove
CDF-S23-001	CDF	\$23	Attachment 3: Sampling Record	Fields		Concentration and Diafiltration Dev Samples DLIMS Project # CoA GxP Samples Project # WHITEFORM GxP Samples Project # Pre-Load Performed By/Date DLIMS Test Name Sample Volume Location Tube Type DLIMS Form Intermediate Storage Temp (oC) DLIMS Storage Temp. (oC) Sample	SMPL: Product Sampling & DLIMS Submission	Cove

						Number -10 (Virus Filtered Product) BIOBURDEN_IP_GPF1 ENDOTOXIN_IP_GPF1 PET -20 (Recovered Product) SOLO_A280 a -30 (Diluted Product, if applicable) SOLO_A280 a -40 (Pre-Formulated Intermediate - Product Quality Samples) SEC_PUR CGE_N_RED SOLO_A280 a ABT Retain c 5 x 14 mL PPS Small Retain b 3 x 14 mL BIOBURDEN_IP_GPF2 ENDOTOXIN_IP_GPF1 PET		
CDF-S23-002	CDF	\$23	Attachment 3: Sampling Record	Instr.	Test SOLO_A280 sample per SOP-0107091. If sample not tested immediately, store sample at 2 – 8°C.		SMPL: Product Sampling & DLIMS Submission	Cove
CDF-S23-003	CDF	\$23	Attachment 3: Sampling Record	Instr.	For PPS Small Retains, select “Retain – PPS Small” for the purpose field when loading into DLIMS.		SMPL: Product Sampling & DLIMS Submission	Cove
CDF-S23-004	CDF	\$23	Attachment 3: Sampling Record	Instr.	For AS Retains, select “Retain – ABT” for the purpose field when loading into DLIMS.		SMPL: Product Sampling & DLIMS Submission	Cove
CDF-S23-005	CDF	\$23	Attachment 3: Sampling Record	Instr.	NOTE: Bolded tests are STAT samples.		SMPL: Product Sampling & DLIMS Submission	Cove
CDF-S24-001	CDF	\$24	Attachment 4: Non-Routine Sampling Record	Note	If non-routine samples are taken, re-adjust weight of the product sampled.		SMPL: Non-Routine Sampling Record	Cove
CDF-S24-002	CDF	\$24	Attachment 4: Non-Routine Sampling Record	Instr.	Log into DLIMS per SOP-0080295 as required by the memo. Record DLIMS Project number and sample number if submitted per DLIMS.	Pre-Load Performed by/Date Process Section No. Contents DLIMS Form DLIMS Project Number Container Type Sample Volume DLIMS Test Name Sample Number Storage Temp. (oC) Location	SMPL: Non-Routine Sampling Record	Cove
CDF-S25-001	CDF	\$25	Attachment 5: Hold Times	Instr.	Record the storage start time, end time, and date for the Virus Filtered Product.		SMPL: Hold Time Table	Cove
CDF-S25-002	CDF	\$25	Attachment 5: Hold Times	Instr.	For product stored at 2 – 8°C, the storage start time is the time when the product is transferred into 2 – 8°C storage, and the storage end time is the time when the product is transferred from 2 – 8°C storage. A new entry for storage at 15 – 25°C begins upon removal from 2 – 8°C.		SMPL: Hold Time Table	Cove
CDF-S25-003	CDF	\$25	Attachment 5: Hold Times	Instr.	Calculate the storage duration for each temperature where the product was stored as well as the combined storage duration, if applicable.	Hold Location Temp. Virus Filtered Product Storage Start (Time/Date) Virus Filtered Product Storage End (Time/Date) Hold Times (hrs and mins) Hold Location Temp. Virus Filtered Product Storage Start (Time/Date) Virus Filtered Product Storage End (Time/Date) R.T. 2-8 °C ____hrs ____mins R.T. 2-8 °C ____hrs ____mins R.T. 2-8 °C ____hrs ____mins Virus Filtered Product Total R. T. Hold Time: If R.T. Total Hold Time > hrs, contact a supervisor Virus Filtered Product Total 2-8 °C Hold Time: Virus	SMPL: Hold Time Table	Cove

						Filtered Product Total 2-8 °C + RT Hold Time: If Total 2-8 °C + RT Hold Time > hrs, contact a supervisor		
CDF-S26-001	CDF	\$26	Attachment 6: Retentate Tank and DF-1244 Set-up	Instr.	(no extractable recordable text)		SMPL: Record CIPDS / Skid / Cassette Information (block stack)	Cove
CDF-S27-001	CDF	\$27	Attachment 7: Product Recirculation Work Sheet	Instr.	Obtain a piece of tubing for recirculation of the product. Record the applicable product name and tubing information. Start mixing the product by recirculation. Record the start time. Mix the product by recirculation per SOP-0107097 for at least 40 minutes and record the mixing information.	Applicable MPR Step Product Name Tubing Part No. Tubing Batch No. Tubing Exp. Date Tubing Autoclave Cycle No. Tubing Autoclave Exp. Date Mixing Start Time Mixing End Time Mixing Duration * (≥ 40 minutes)	SMPL: Product Recirculation Worksheet	Cove
CDF-S27-002	CDF	\$27	Attachment 7: Product Recirculation Work Sheet	Instr.	* The duration of mixing is the time that elapses between the mixing end time and the mixing start time in minutes.		SMPL: Product Recirculation Worksheet	Cove
CDF-S28-001	CDF	\$28	Attachment 8: Yield Calculations	Instr.	Final Product Mass, Pre-Sampling	PFI Volume before sampling Final Product Concentration Final Product Mass, Pre-Sampling (a) _____L_ (Step 17.16 or Step 16.33 if no dilution) _____g/L (8012475-20 or 8012475-30 *) Volume of the PFI after sampling and after transfer Final Product Concentration Final Product Mass, Post-Sampling (b) _____L_ (Step 18.9) _____g/L (8012475-20 or 8012475-30 *)	SMPL: Yield Calculations	Cove
CDF-S28-002	CDF	\$28	Attachment 8: Yield Calculations	Instr.	* Use (-30) if product was diluted.	Final Product Mass, Pre-Sampling (a) Total Grams of Virus Filtration Product after sampling (Step 9.10) Yield % Step Yield Final Product Mass, Post-Sampling (b) Total grams of Virus Filtration Product before sampling (Step 7.8) Yield % MPR Yield	SMPL: Yield Calculations	Cove
CDF-S29-001	CDF	\$29	Attachment 9: Comments Section	Fields		Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
CDF-S29-002	CDF	\$29	Attachment 9: Comments Section	Instr.	Attachment 9: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
CDF-S29-003	CDF	\$29	Attachment 9: Comments Section	Instr.	Attachment 9: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
CDF-S29-004	CDF	\$29	Attachment 9: Comments Section	Instr.	Attachment 9: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
CDF-S29-005	CDF	\$29	Attachment 9: Comments Section	Instr.	Attachment 9: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove
CDF-S29-006	CDF	\$29	Attachment 9: Comments Section	Instr.	Attachment 9: Comments Section (continued)	Step No. Comments Please initial and date each entry. Initials/ Date	SMPL: Comments / Deviations Section	Cove

CDF-S30-001	CDF	\$30	MASTER PRODUCTION RECORD REVISION HISTORY	Instr.	The following is included for reference only and is not part of the executable record.	Version Change Reason This is a new master production record (MPR)	— (revision history) —	Excl
CDF-S30-002	CDF	\$30	MASTER PRODUCTION RECORD REVISION HISTORY	Instr.	The information contained in this document is derived from Technical report and sampling and testing plan PLAN-0124866. Additional information may be found in REP-0414101.		— (revision history) —	Excl

4 · Streamlined-Analysis reconciliation (per streamlined step)

Rec	Streamlined	Streamlined Title	Fulfilled by XStep(s)	Status	Note / Action
VI	7.1	Measure pH, Conductivity, Temperature	Common - Product pH Conductivity Temperature	Covered	
VI	7.14	Starting Product Temperature Check / Decision Tree	VI - Temperature Decision Tree	Covered	
VI	8.4	VI Treatment Vessel Setup - Comprehensive	VI - Treatment Vessel Setup	Covered	
VI	10.9	Acid Treatment - Iterative pH Titration Table	VI - Acid-Base pH Titration Table	Covered	
VI	12.1	Ensure Agitation (Extension)	Common - Product Mixing	Covered	
VI	14.11	Neutralization Treatment - Iterative pH Titration	VI - Acid-Base pH Titration Table	Covered	Same reusable XStep as 10.9 (acid/base variant).
VI	14.15	Calculate Net Weight of Neutralized VI Product	Common - Calc Three Columns	Covered	
VI	15.5	Measure Final pH, Conductivity, Temperature	Common - Product pH Conductivity Temperature	Covered	
VI	15.8	Store Product and Document Hold Times	VI - Store Product and Hold-Time Link	Covered	
VI	ATT-3	Attachment 3: Hold Times	Common - Hold Time Table	Covered	Storage linkage also on VI - Store Product.
VI	ATT-5	Attachment 5: Transfer Tubing/Hosing Information	Common - Transfer Tubing and Hosing Info	Covered	
C&D	7.2	Record CIPDS/Skid Information	CDF - Record CIPDS Skid Cassette Info	Covered	
C&D	7.4	Record VF Product Information	Common - Incoming Product Information	Covered	Now covered by the reusable Incoming Product Information XStep (tare/net weight + concentration + DLIMS) — added during the \$7.5 review.
C&D	7.6	Calculate Total Grams VF Product per Vessel	Common - Calc Three Columns	Covered	
C&D	7.7	Calculate Minimum Membrane Surface Area	CDF - Minimum Membrane Surface Area	Covered	
C&D	8.12	Sample Retentate/Permeate — Record Results	Common - Product Sampling and DLIMS Submission + Common - Product pH Conductivity Temperature	Covered	
C&D	9.2	Mix VF Product (>=15 min)	Common - Product Mixing	Covered	
C&D	9.8	Weigh Product Vessels — Calculate Net Weight	Common - Product Vessel Weigh	Covered	

C&D	11.1	Obtain & Label Retentate Vessel (TK-1927)	CDF - Record CIPDS Skid Cassette Info	Folded	§11 Vessel Set-up folded here; consider a dedicated "Obtain & Label Vessel" XStep.
C&D	11.6	Record Tare Weight Before Attachment	Common - Product Vessel Weigh	Covered	
C&D	14.11	Sample Retentate/Permeate (pH/Cond/Temp)	CDF - Continued Diafiltration Decision	Covered	Sampling + Measure blocks also apply.
C&D	15.6	Record Conc 2 End Time/Weight/Volume	CDF - Run Skid Recipe + Common - Product Vessel Weigh	Covered	
C&D	16.34	Calculate Recovery Product Volume	CDF - Recovery Calculations and Execution	Covered	
C&D	18.1	Obtain PFI Storage Vessel	CDF - PFI Storage Vessel and Filtration	Covered	
C&D	18.4	Obtain SHC Filter	CDF - PFI Storage Vessel and Filtration	Covered	
C&D	ATT-2	Operation Parameters	CDF - Operation Monitoring Worksheet	Covered	
C&D	ATT-7	Product Recirculation Work Sheet	Common - Product Recirculation Worksheet	Covered	
C&D	ATT-8	Yield Calculations	Common - Yield Calculations	Covered	
C&D	--	SOLO_AZ280 (SoloVPE concentration determination)	Common - Incoming Product Information (result) + Common - Product Sampling (submission)	Covered	REVISED after the §7.5 review: the SoloVPE concentration RESULT (g/L) IS recorded in the BR (VI §7.5 "Affinity Product SoloVPE Concentration", C&D §7) — captured by the Incoming Product Information XStep with its DLIMS project/sample numbers. Sample submission is the Sampling Record XStep; the determination itself is the external SoloVPE instrument op (SOP-0107091). No dedicated SoloVPE worksheet XStep needed.
C&D	--	Recirculation Protocol	Common - Product Recirculation Worksheet	Review	May be a standalone protocol rather than an XStep — confirm scope.
C&D	--	Cassette Installation	Separate form FORM-0071789	Out-of-scope	C&D §7: "Obtain and record ... from the AZD0543 CIPDS (FORM-0071789)". Cassette install + integrity is on its own data sheet; only summary info is transcribed into §7 (covered by CDF - Record CIPDS Skid Cassette Info).
C&D	--	VPro test work sheet (filter integrity)	Separate record MABR-0027575 / PN 8012474	Out-of-scope	RESOLVED: "VPro" = the Virus Filtration (nanofiltration) record MABR-0027575 / PN 8012474 — a SEPARATE MPR not in our scope. UF/DF cassette integrity is on FORM-0071789. "Filter Integrity Tester" appears only as a BOM equipment line; no integrity procedure exists in either BR body. NO new XStep needed.

5 · Goods Issue & Label points (17 GI · 7 label)

From a keyword sweep of both MPRs for "SAP Consumption" (Goods Issue, Z_PICONS mvt 261) and "label" / SOP-0107056. Rows marked **Yes (added)** are gaps this review found and closed in the XStep instructions. **Open question:** buffer 8012555 Goods Issue — one consolidated posting or per recipe phase?

Rec	§	Type	Material / Label target	Fulfilled by XStep	Reflected?
VI	§4	GI	Bill of Materials components (SAP Consumption)	Common - Display BOM	Yes
VI	§8	GI	VI Treatment Vessel components	VI - Treatment Vessel Setup	Yes
VI	§8	LABEL	VI Treatment Vessel (AZD0543 / batch / part / initials / date)	VI - Treatment Vessel Setup	Yes (added)
VI	§10	GI	1M Acetic Acid (A-005168)	VI - Acid-Base pH Titration Table	Yes

VI	\$12	GI	1M Acetic Acid (A-005168) - extension	VI - Acid-Base pH Titration Table	Yes
VI	\$14	GI	1M Tris Base (A0101-D)	VI - Acid-Base pH Titration Table	Yes
VI	\$7/\$15	LABEL	Product samples (per SOP-0107056)	Common - Product Sampling and DLIMS Submission	Yes
CDF	\$4	GI	Bill of Materials components	Common - Display BOM	Yes
CDF	\$7	GI	CIPDS / skid material consumption	CDF - Minimum Membrane Surface Area	Yes (added)
CDF	\$8	GI	WFI (pre-use flush)	CDF - Run Skid Recipe	Yes (added)
CDF	\$10	GI	Equilibration buffer PN 8012555	CDF - Run Skid Recipe	Yes (added)
CDF	\$11	GI	Retentate Vessel / DF-1244 components	CDF - Record CIPDS Skid Cassette Info	Yes (added)
CDF	\$11	LABEL	Retentate Vessel + manual valves A-I (per SOP-0107056)	CDF - Record CIPDS Skid Cassette Info	Yes (added)
CDF	\$13	GI	Concentration 1 buffer PN 8012555	CDF - Run Skid Recipe	Yes (added)
CDF	\$14	GI	Diafiltration buffer PN 8012555	CDF - Continued Diafiltration Decision	Yes (added)
CDF	\$15	GI	Concentration 2 (recipe)	CDF - Run Skid Recipe	Yes (added)
CDF	\$16	GI	Recovery buffer PN 8012555	CDF - Recovery Calculations and Execution	Yes
CDF	\$16	LABEL	PFI mixing bag (per SOP-0107056)	CDF - Recovery Calculations and Execution	Yes (added)
CDF	\$17	GI	Dilution buffer PN 8012555	CDF - Dilution Decision and Calculations	Yes
CDF	\$17	LABEL	"Intermediate Vessel" + sample (per SOP-0107056)	CDF - Dilution Decision and Calculations	Yes (added)
CDF	\$18	GI	SHC filter / storage bag	CDF - PFI Storage Vessel and Filtration	Yes
CDF	\$18	LABEL	PFI storage bag / bottle (per SOP-0107056)	CDF - PFI Storage Vessel and Filtration	Yes
CDF	\$19	GI	NaOH (A0014-D) post-use sanitization	CDF - Post-Processing and Sanitization	Yes
CDF	\$23	LABEL	Retentate / permeate samples (per SOP-0107056)	Common - Product Sampling and DLIMS Submission	Yes

6 · XStep inventory (32)

#	Group	XStep (SMPL)	Old BR sections fulfilled	# Old groups
1	Common	SMPL: Order Header Details	(reusable — instantiated in-line)	0
2	Common	SMPL: Signature Table	(reusable — instantiated in-line)	0
3	Common	SMPL: Display BOM	CDF \$4, VI \$4	15
4	Common	SMPL: Room & Equipment Assign	(reusable — instantiated in-line)	0
5	Common	SMPL: Additional Manufacturing Supplies	(reusable — instantiated in-line)	0
6	Common	SMPL: Additional Solution Batches (Solution Switch)	CDF \$5, VI \$5	13
7	Common	SMPL: Process Notes & Global Limits (Acknowledgement)	CDF \$6, VI \$6	16

8	Common	SMPL: Incoming Product Information (Weight / Concentration / DLIMS)	VI §7	17
9	Common	SMPL: Product Vessel Weigh	(reusable — instantiated in-line)	0
10	Common	SMPL: Calc — Three Columns (Value1 op Value2 = Result)	(reusable — instantiated in-line)	0
11	Common	SMPL: Multi-Variable Calculation (3-input / ranged)	(reusable — instantiated in-line)	0
12	Common	SMPL: Product Mixing	CDF §9	10
13	Common	SMPL: Product Sampling & DLIMS Submission	CDF §23, VI §15, VI §17	21
14	Common	SMPL: Product pH / Conductivity / Temperature	(reusable — instantiated in-line)	0
15	Common	SMPL: Hold Time Table	CDF §25, VI §19	6
16	Common	SMPL: Product Recirculation Worksheet	CDF §27, VI §22	4
17	Common	SMPL: Transfer Tubing / Hosing Information	CDF §21, VI §21	5
18	Common	SMPL: Product Storage & Cross-Record	(reusable — instantiated in-line)	0
19	Common	SMPL: Instruction & Sign-off	CDF §20, VI §16	8
20	Common	SMPL: Yield Calculations	CDF §28, VI §20	4
21	Common	SMPL: Comments / Deviations Section	CDF §29, VI §23	10
22	Common	SMPL: Non-Routine Sampling Record	CDF §24, VI §18	4
23	VI	SMPL: VI Treatment Vessel Setup	VI §8	13
24	VI	SMPL: Iterative pH Titration Table (Acid / Base)	VI §10, VI §12, VI §14	46
25	VI	SMPL: Incubation Timing & Incubated Sample (block stack)	VI §11, VI §13	9
26	VI	SMPL: Starting / Transfer Temperature Decision (block stack)	VI §9	7
27	VI	SMPL: Store VI Product & Hold-Time Link (block stack)	(reusable — instantiated in-line)	0
28	CDF	SMPL: Record CIPDS / Skid / Cassette Information (block stack)	CDF §11, CDF §26	11
29	CDF	SMPL: Run Skid Recipe (block stack)	CDF §8, CDF §10, CDF §13, CDF §15	88
30	CDF	SMPL: Minimum Membrane Surface Area (block stack)	CDF §7	8
31	CDF	SMPL: Process Input Calculations (calc-block stack)	CDF §12	9
32	CDF	SMPL: UF/DF Operation Monitoring Worksheet (Att 2)	CDF §22	16
33	CDF	SMPL: Conditional Continued Diafiltration (block stack)	CDF §14	27
34	CDF	SMPL: Recovery Calculations & Execution (block stack)	CDF §16	33
35	CDF	SMPL: Dilution Decision & Calculations (block stack)	CDF §17	35
36	CDF	SMPL: PFI Storage Vessel, SHC Filter & Store (block stack)	CDF §18	13
37	CDF	SMPL: Post-Processing: Cassette Re-use & Sanitization (block stack)	CDF §19	15